Fractal Toroidal-Beat / QHD Plasmoid: A Tiered Synthesis

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A Driven Beltrami-Hopf Plasmoid as One Coherent Object Across Scales: A Tiered Synthesis

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Abstract

This paper describes a single physical object – one driven, force-free Beltrami-Hopf standing wave, a Chandrasekhar-Kendall (CK) ball mode satisfying `curl B = lambda B`, read at once as a plasma configuration and as a Madelung / quantum-hydrodynamic (QHD) matter wave – and shows how mass, spin, charge, topology, a living beat rhythm, and the low-energy-nuclear (LENR / CMNS) anomaly all follow from a single organizing law. That law, the *spine*, is that the Madelung continuity of the matter-wave fluid **is** the topological conservation of the object's magnetic helicity, joined by a profile proportionality `rho = kappa (A.B)` in which `kappa` is a fixed dimensional constant and `A.B` is a helicity density. The defining device of the paper is that every key law is written in two columns – REAL-UNIT (dimensional) beside DIMENSIONLESS (Buckingham-Pi group) – each carrying a claim -> Pi-group -> falsifier. Mass is a rate, `m = hbar omega_C / c^2`, dimensionally exact; the whirl *number* `q = 1/alpha ~ 137` is dimensionless and kept strictly separate. The topology carries a *real* helicity `H ~ 0.088`, not an integer Hopf charge. The object is *alive*: a Stuart-Landau limit cycle whose family self-organizes by Kuramoto/Adler/Arnold dynamics, breathing through a twist-writhe heartbeat that conserves the Calugareanu-White linking `Lk = Tw + Wr`.

Two results define the current version of the theory. First, the one residual left open in the organizing law – the *current leg* of the four-current identification – is now resolved as a trilogy: a verified no-go in the static regime, a verified no-go in the aligned-steady regime, and a constructed, realizable closure in the genuinely driven regime, with the object's own dynamics shown to *select* the no-go (closure is externally driven, not emergent). Second, at the nuclear scale the result is *scaffold-not-reaction*: the object supplies the coherent environment, the cold formation, the merger orientation, and the disposal character, while the anomaly reduces to two static inputs – an internal branching amplitude and an inherited, literature-measured electron screening – with no over-unity (a coefficient of performance `COP ~ 1.3-1.4` is inherited field positioning, not a derived or claimed excess).

Everything traces to four measured anchors `{B, n_i, m_i, R}` with **zero free structural parameters**. The work is presented as a tiered SYNTHESIS HYPOTHESIS, with its load-bearing machinery credited to established, peer-reviewed physics and its open frontiers reduced to named external computations – not as a claimed-proven grand unification.

Scope: what is and is not claimed

Because this paper draws one picture across plasma physics, quantum hydrodynamics, topology, and low-energy nuclear phenomenology, it is essential to state at the outset exactly what is being asserted and at what confidence. This is a *synthesis hypothesis*, tiered claim by claim; it is not a claimed-proven theory of everything.

What is claimed. (1) That a single coherent object – a driven, force-free Beltrami-Hopf plasmoid – can be read consistently across scales as electron, nucleon, exotic vacuum object (EVO), and neutrino, differing by scale and boundary condition rather than by kind. (2) That one organizing law, the profile proportionality `rho = kappa (A.B)`, ties the matter-wave continuity of the object to the conservation of its magnetic helicity, with a verified density leg and a current leg that is now named and resolved. (3) That mass is a dimensionally exact rewriting as a rate, `m = hbar omega_C / c^2`. (4) That the particle mass ladder admits a spectral-geometry reading on nested Hopf shells. (5) That the object is a living, driven limit cycle whose family self-organizes by standard synchronization dynamics. (6) That the LENR / CMNS anomaly reduces, under this reading, to a *scaffold-not-reaction* boundary in which the object supplies the environment and inherits a well-posed nuclear-structure kernel. Every quantitative value is built from four anchors with zero free structural parameters.

What is not claimed. This work does **not** claim a proven grand unification; a new fundamental force; over-unity, free energy, or any device-level excess power; a first-principles *derived value* of the fine-structure constant `alpha`; derived neutrino *mass values*; or a closed LENR reaction *rate*. Where the theory reaches a limit, that limit is reported as *our* computational limit on *our* object, and – in the deepest cases – as a named external computation (a 3D global-regularity bound, an external Majorana seesaw sector, a compiled topology-preserving Skyrme relaxation), not as a verdict against the Standard Model, classical electromagnetism, or any collaborator's experimental program. The object-reading sits *atop* the incumbent physics; it does not overturn it.

Support-lens framing. The scope of the object is the driven charge-cluster / EVO of the Greenyer-Shoulders lineage, not universal ball lightning. Statements about the work of other investigators are supportive: our limits are ours to own.

Tier legend

Every nontrivial statement in this paper carries exactly one tier tag, one convention throughout. The legend is the paper's honesty instrument; it is stated once here and never relaxed.

- **[V]** – *verified*: established by a named computation or identity performed and checked in this work (residuals quoted where numerical).
- **[credited]** – *established physics built upon*: a peer-reviewed result used as given, cited to its source. Load-bearing machinery is always credited.
- **[S]** – *structural*: worked on our side but not closed; a construction whose scaffolding is in place but whose final value or proof is outstanding.
- **open** – a named, still-open problem or an internal quantity not yet closed; where deepest, reduced to a named external computation. (Retires the former bare `[A]` tag.)
- **[postulate]** – a consistent defining assumption, declared as such.
- **[QWM framework]** – a refinement in the Reed / quantum-wave-mechanics reading. It is *corroborating and never load-bearing*: the four anchors, not the QWM picture, set every number.
- **[TUFT-preprint]** – material read from Nielsen's *Toroidal Unified Field Theory* preprint, which is *in peer review* – neither peer-reviewed nor self-published. Marked wherever it is load-bearing; never presented as established.
- **[analogy]** – an explicit modeling analogy, never an identity.
- **[clue]** – a near-coincidence held open and pursued for a mechanism, never promoted on the numerical digits alone.

- **NEGATIVE** – a first-class negative result: specific, computed, and tested. A NEGATIVE is a finding, not a category dismissal, and is reported with the same standing as a positive.
- [**do-not-cite**] – material deliberately excluded from the load-bearing science.

Mixed tags (e.g. `[V grade / S value]`) mean the *structure* of a claim is verified while its *value* remains structural. Throughout, `mu_0` and `hbar`, `c` are universal constants, not anchors.

Glossary and concepts

Every term is defined here, in plain physical English, before it is used in the body. Read this section first; the rest of the paper leans on it.

The object and its two readings

- **The object (FTGB plasmoid).** One coherent, self-trapped standing wave – a small, driven, toroidal plasma structure of the kind reported as an "exotic vacuum object" (EVO) or charge cluster. Throughout, "the object" means this single thing, read two ways at once.
- **Force-free (Beltrami) field.** A magnetic field whose electric currents run everywhere *parallel* to the field they carry, so the Lorentz force `j x B` vanishes in the bulk. The field no longer pushes on its own currents. Mathematically `curl B = lambda B`.
- **Chandrasekhar-Kendall (CK) mode.** The lowest force-free eigenfield of the curl operator confined to a ball. This is the concrete mathematical shape of the object as a plasma.
- **Woltjer-Taylor relaxation.** The physical process by which a turbulent magnetized plasma settles into the lowest-energy force-free state at fixed helicity. It means the object is an *attractor* – the state a real plasma falls into – not a fine-tuned initial condition.
- **Madelung / quantum-hydrodynamic (QHD) reading.** The reading of a quantum wavefunction `psi = sqrt(rho) exp(iS/hbar)` as a real, compressible *fluid* rather than only a probability amplitude. Substituting this form into the Schrodinger equation splits it *exactly* into a continuity equation for the density `rho = |psi|^2` flowing with velocity `v = grad(S)/m`, plus a Hamilton-Jacobi equation carrying the Bohm quantum potential. The continuity half is exactly `d_t rho + div(rho v) = 0`. We use the flow reading because, under it, the plasma law and the matter-wave law become the *same* equation.
- **Bohm quantum potential `Q`.** The internal pressure term the Madelung split produces. It is what lets the wave *stand* rather than disperse – the reason the object is a soliton (a self-trapped standing matter wave) and not a spreading wave packet.
- **Polarizable physical vacuum.** The substrate the wave stands in and moves through: a genuine dielectric of relative permittivity `eps_r -> 1`, modeled as a fine-scale medium of dipoles. Its fundamental oscillator quantum sets the object's absolute energy scale.

The organizing quantities

- **Magnetic helicity `H = integral A.B dV`.** A single number measuring how much the field lines wrap around and link one another; `A` is the magnetic vector potential, `B = curl A`. It is a conserved, gauge-invariant topological invariant. In plain terms, it counts "how many times these nested rings thread each other."
- **Helicity density `K^0 = A.B`.** The local density whose volume integral is `H`. For a force-free field in the natural gauge `A = B/lambda` it is `|B|^2/lambda >= 0` – one-signed, and equal to the pointwise squared *magnitude* of the field divided by the eigenvalue.
- **Canonical (generalized) helicity.** The two-fluid generalization of magnetic helicity, `integral (A + (m/q) v) . (B + (m/q) curl v) dV`, which is a Casimir invariant of the ideal two-fluid plasma. It is the conserved current that carries the *driven* current leg (Part D); its ideal conservation is the field's, not ours [credited: Steinhauer-Ishida 1997; Mahajan-Yoshida 1998].
- **The spine (density-matter law) `rho = kappa (A.B)`.** The central claim: matter density and helicity density share **one profile**, tied by a fixed *dimensional* constant `kappa`. Because `kappa` is a constant

it slides out of both divergences, so the matter-wave continuity equation and the helicity-conservation equation become the same statement. `A.B` is a helicity density (units of `T^2 . m`), not a matter density (`kg/m^3`); the two are related by a proportionality, never by a raw equality.

- **Anchors `{B, n_i, m_i, R}`.** The four measured inputs – medium field strength, ion number density, ion mass, and object radius – from which every quantitative value is built, with **zero free structural parameters**.
- **Alfven speed `v_A = B/sqrt(mu_0 n_i m_i)`.** The one velocity buildable from the anchors and the universal constant `mu_0`. With the radius `R` it gives the one frequency scale, the transit rate `f_A = v_A/(2 pi R)`. Every frequency the object carries is `f = Lambda . f_A` for some dimensionless eigenvalue `Lambda`.

Mass, charge, and the whirl

- **Whirl / Compton frequency `omega_C`.** The angular rate at which the phase of the confined, circulating matter wave advances. A heavier particle is one whose internal wave winds its phase faster.
- **Mass as a rate `m = hbar omega_C / c^2`.** Mass is the *energy of a rate* – the whirl frequency read as an energy through the de Broglie / Compton identity. Mass carries kilograms; it is a function of the whirl frequency, not identical to any pure number.
- **Zitterbewegung `omega_zbw = 2 m c^2 / hbar`.** The "trembling" clock of the localized matter wave. The mass axis rides at exactly half this rate (`omega_C = omega_zbw/2`), which is why spin-1/2 appears as a clean 2:1 rotary octave.
- **Whirl number `q = 1/alpha = 137.036`.** A *dimensionless* count: spin revolutions per closed orbital whirl. For the electron this is about 137, and the electromagnetic fine-structure constant is read as its inverse. The whirl number is a separate quantity from the mass; no equation slides between a mass in kilograms and a pure ratio.
- **Charge as spin angular momentum, `e ~ [kg . rad/s]`.** Electric charge read as an intrinsic, Lorentz-invariant mechanical angular momentum, so the Coulomb becomes a *count* of electrons rather than a distinct kind of stuff. Geometrically, charge is a *loop-closure torsion defect*: the precessing spin winding fails to close and re-synchronizes about every 137 turns.

Topology and spectral geometry

- **Hopf fibration `S^1 -> S^3 -> S^2`.** The classic map in which every pair of fibre circles on the three-sphere links exactly once. Helicity is the physical realization of this linking count (the `pi_3(S^2)` invariant).
- **Real helicity vs integer charge.** For the actual CK object the winding is a genuine conserved *real* number `H ~ 0.088`, not an integer topological charge. The integer picture is an idealization used for closed-fibre reference fields (the neutrino smoke-ring, the idealized toroidal electron).
- **Baryon number `B` (Skyrme degree).** The integer `pi_3(S^3)` winding of a soliton's flux field – a *different* topological reading on a *different* target space from helicity. Mass and baryon number are two independent readings of the same object, never slid between.
- **The four torsions.** Four genuinely distinct objects that share the word "torsion," kept rigidly apart: (a) Cartan/Burgers loop-closure torsion (behind charge), (b) ribbon twist `Tw` (the reconnection heartbeat), (c) Frenet-Serret torsion (a chirality signature), (d) Ray-Singer analytic torsion (the mass-tower torsion coefficient). Einstein-Cartan spacetime torsion sits ~15 orders below the Compton scale and is excluded as dynamically irrelevant.
- **Mass tower / spectral geometry.** The particle mass ladder read as the eigenvalue spectrum of a geometric operator (`*d`, Hodge-star composed with exterior derivative) on nested Hopf shells `S^3, S^5, S^7, S^9`. The tower's quadratic Casimir sector is pi-free; its torsion sector carries the `1/pi^2`.

The living dynamics

- **Stuart-Landau limit cycle.** The universal equation for a self-oscillator sitting just past a Hopf bifurcation. Its amplitude locks onto a ring of fixed radius `r*` – the object breathes at that radius rather than growing or decaying.

- **Heartbeat (twist-writhe exchange).** The physical realization of the limit cycle: helicity shuttles between internal ribbon twist `Tw` and global writhe `Wr` once per closure, conserving the Calugareanu-White linking `Lk = Tw + Wr`. This is a renewed rhythm (a Prigogine dissipative structure), not a stored-spin flywheel, and it carries no over-unity.
- **Carrier comb.** The object's resonant frequencies, `f_n = Lambda_n v_A/(2 pi R) = {121, 208, 294} kHz`, an *inharmonic* comb in ratios `1 : 1.72 : 2.43` set by the CK Bessel roots.
- **Kuramoto / Adler / Arnold synchronization.** The standard, textbook machinery of coupled oscillators, governing when a *family* of these living objects locks onto a shared rhythm and when it stays incoherent. Controlled by one dimensionless parameter, `P = K/K_c`.

The nuclear boundary

- **Scaffold-not-reaction.** The paper's nuclear headline: the object *supplies* the coherent environment, the cold formation, the merger orientation, and the disposal character of the LENR / CMNS anomaly, and *inherits* exactly one input – the sub-barrier penetration kernel.
- **The kernel / branching amplitude `M_fi`.** The nuclear-structure matrix element `M_fi = <Psi_f | O | Psi_i>` governing which exit channel the reaction takes. It is a well-posed, published-grounded open computation, not a missing principle.
- **The scalar `S = mu - P.v`.** The scalar field whose vanishing is the closure condition of the driven current leg (Part D): `mu` is the chemical/enthalpy potential and `P.v` the pressure-flow work density. Closure of the current leg holds if and only if `S = 0` on a non-empty, index-1, bounded surface.
- **The two static inputs `{Delta, U_s}`.** The two numbers the LENR boundary reduces to, with *different owners*: **`Delta`**, the internal branching amplitude (FTGB-internal, open [S]); and **`U_s`**, the inherited metallic electron-screening potential, `U_s ~ 300-800 eV`, taken from the literature (accepted-not-adjudicated), not derived from the object.

Claim tiers

Every nontrivial statement carries one tier tag per the Tier legend above ([V] / [credited] / [S] / open / [postulate] / [QWM framework] / [TUFT-preprint] / [analogy] / [clue] / NEGATIVE / [do-not-cite]).

Part A – The established-physics foundation (foregrounded)

A.1 Introduction: one object, two columns, and the load-bearing physics

This paper is about a single physical thing described two ways at once, and about a single method for keeping that description honest. Before any FTGB-specific claim is made, this part does one deliberate thing: it foregrounds the peer-reviewed physics the whole edifice rests on. The originality of this work is a *synthesis*, not a new force; the machinery that carries the load – force-free fields, helicity conservation, the hydrodynamic form of quantum mechanics, topological solitons, and mass as a frequency – is standing, established physics, and it is credited as such here so that the reader can separate what is built *on* from what is *ours*.

The object. As a *plasma* it is a driven, force-free Chandrasekhar-Kendall mode – the lowest Beltrami eigenfield of the curl operator confined to a ball, `curl B = lambda B`, whose currents run parallel to the field so the Lorentz force vanishes in the bulk. It is the minimum-energy state that Woltjer-Taylor relaxation selects at fixed helicity: an attractor, not a fine-tuned condition. As a *matter wave* the same field is a Madelung (quantum-hydrodynamic) fluid, the wavefunction read as a real compressible flow. We read the object through this flow ontology because, under it, the plasma law and the matter-wave law become the *same* equation. Later parts read mass, spin, charge, and topology off this one confined flow; show that the object is alive; and carry the law to the nuclear scale, where it becomes a sharp, decidable boundary rather than a new force.

The method. Every key law is displayed in two columns: the REAL-UNIT (dimensional) form – the physics as a plasma physicist or a Madelung theorist would ordinarily write it – beside the DIMENSIONLESS (Buckingham-Pi group) form, the number of order one an experiment can test directly [V-structure / credited: Buckingham 1914]. Each pairing carries a three-part framing: a **CLAIM** (the physical statement), its **Pi** (the dimensionless content the theorem can fix), and a **FALSIFIER** (the observation that would kill it). Buckingham-Pi is silent about the pure *numbers* – eigenvalues, coefficients, correlations – which come from geometry, calibration, or measurement; the side-by-side format makes that division of labor explicit. Every dimensional prefactor (`kappa` first, others later) is a constant the Pi theorem cannot fix and that must be pinned by mechanism or data.

The discipline. Everything traces to four anchors `{B, n_i, m_i, R}` with **zero free structural parameters**. Claims are tiered inline (see the Tier legend). `mu_0` is a universal constant, not an anchor.

What we build on, and what is ours. It is worth stating the division of labor plainly, because it governs how every later claim should be read. *Built on (peer-reviewed, credited):* the force-free Beltrami / CK eigenmode construction and its ball spectrum; magnetic helicity as a conserved, gauge-invariant topological invariant, and Woltjer-Taylor relaxation as the attractor that selects it; the exact Madelung rewriting of Schrodinger flow and the Bohm quantum potential; the Skyrme topological-soliton picture with baryon number as the `pi_3(S^3)` degree; the de Broglie / zitterbewegung internal clock and mass as a frequency; the Stuart-Landau amplitude equation and the Kuramoto/Adler/Arnold synchronization family; and the two-fluid (canonical) helicity Casimir. *Ours (the synthesis):* the identification of these as *one object read across scales*; the spine `rho = kappa (A.B)` tying matter continuity to helicity conservation through a single dimensional constant; the reading of mass, spin, charge, and topology off that one confined flow; the current-leg trilogy; and the scaffold-not-reaction LENR boundary. Nielsen's TUFT preprint (mass-tower spectral geometry) and Reed's QWM framework (charge as spin angular momentum, the whirl number) are used only as *marked-tiered* material – [TUFT-preprint] and [QWM framework] respectively – never as established physics, and the four anchors, not those pictures, set every number.

A.2 The foundation, stated as standing physics

This section gathers the five pillars the whole theory rests on and states each *as established physics, credited before it is used*. Nothing in this section is an FTGB claim; it is the bedrock. The FTGB-specific construction begins in Part B and refers back here.

A.2.1 Force-free (Beltrami) fields and the Chandrasekhar-Kendall ball mode

A *force-free* magnetic field is one in which the current density runs everywhere parallel to the field, so that `j x B = 0` in the bulk and the field exerts no net Lorentz force on the plasma that carries it. The defining equation is the Beltrami condition

```
curl B = lambda B ,
```

an eigenvalue problem for the curl operator. Chandrasekhar and Kendall solved this problem explicitly for a field confined to a sphere, constructing the complete set of force-free eigenmodes now standard in the field [credited: Chandrasekhar & Kendall 1957, *Astrophys. J.* **126**, 457]. The boundary condition of vanishing normal field on a conducting ball, `B_r(R) = 0`, quantizes the eigenvalue: only discrete `lambda` fit. The lowest nonzero root of the spherical-Bessel condition `j_1(lambda R) = 0` gives the ground-state eigenvalue `lambda_1 R = 4.4934` (the first nonzero root of `tan x = x`), and the higher roots `{4.4934, 7.7253, 10.9041, ...}` form an *inharmonic* comb in ratios `1 : 1.719 : 2.427 : ...`. This is entirely established plasma physics; the FTGB use of it (as the plasma face of the object, and as the origin of the carrier comb) is deferred to Parts B and C.

That force-free states are not fine-tuned but *selected* is the second established fact. Woltjer proved that the force-free configuration is the minimum-magnetic-energy state at fixed helicity [credited: Woltjer 1958, *Proc. Natl. Acad. Sci. USA* **44**, 489], and Taylor showed that a turbulent, weakly resistive plasma relaxes toward the single-`lambda` Beltrami state by selective decay of energy at nearly fixed helicity [credited: Taylor 1974, *Phys. Rev. Lett.* **33**, 1139]. Taylor relaxation is why a real plasma *falls into* the force-free state rather than requiring it as an initial condition. The geometric underpinning is equally standard: the Beltrami field is exactly the Reeb field of a contact structure whose characteristic foliation is the Hopf fibration [credited: Etnyre & Ghrist 2000, *Nonlinearity* **13**, 441], which is what makes the winding of a force-free field a genuine topological quantity rather than an incidental geometric feature.

A.2.2 Magnetic helicity as a conserved topological invariant

The conserved quantity of an ideal force-free field is its **magnetic helicity**,

```
H = integral A.B dV ,     B = curl A ,
```

a single number that measures how much the field lines wrap around and link one another. Helicity is gauge-invariant for a field with no normal component on the boundary, and it is conserved under ideal (non-resistive) evolution. Its topological meaning is the linking of field lines: for two linked flux tubes the helicity equals the product of their fluxes times the (integer) linking number. This interpretation of helicity as a conserved topological invariant of the field is due to Moffatt [credited: Moffatt 1969, *J. Fluid Mech.* **35**, 117], building on Woltjer's minimum-energy result. Helicity realizes the `pi_3(S^2)` (Hopf) invariant: for a clean, closed-fibre idealization it counts the linking of the circular fibres of the Hopf fibration `S^1 -> S^3 -> S^2`, in which every pair of fibres links exactly once.

The ribbon decomposition of helicity into internal twist and global writhe is likewise standard: the Calugareanu-White identity `Lk = Tw + Wr` conserves the linking of a flux ribbon while allowing twist and writhe to exchange, which is the established bookkeeping behind the reconnection dynamics used later [credited: Moffatt & Ricca 1992, *Proc. R. Soc. A* **439**, 411]. In the two-fluid setting, magnetic helicity generalizes to *canonical (generalized) helicity*, a Casimir invariant of the ideal two-fluid plasma; the double-Beltrami relaxed states that carry it are established results [credited: Steinhauer & Ishida 1997, *Phys. Rev. Lett.* **79**, 3423; Mahajan & Yoshida 1998, *Phys. Rev. Lett.* **81**, 4863]. Part D leans on this generalized invariant, not on any new conservation law of ours.

A.2.3 The Madelung / quantum-hydrodynamic form of quantum mechanics, and the Bohm potential

Quantum mechanics can be rewritten *exactly* as the dynamics of a compressible fluid. Writing the wavefunction in polar form,

```
psi = sqrt(rho) . exp( i S / hbar ) ,
```

and substituting into the time-dependent Schrodinger equation splits it, with no approximation, into two real equations: a continuity equation for the density `rho = |psi|^2` flowing with velocity `v = grad(S)/m`,

```
d_t rho + div(rho v) = 0 ,
```

and a quantum Hamilton-Jacobi equation for the phase `S`, which differs from its classical counterpart by a single additional term, the **Bohm quantum potential**

```
Q = -(hbar^2 / 2m) . grad^2( sqrt(rho) ) / sqrt(rho) .
```

The continuity half is *exactly* the classical continuity equation; the whole quantum content of the flow is carried by `Q`. This hydrodynamic reformulation is due to Madelung [credited: Madelung 1927, *Z. Phys.* **40**, 322], and the quantum potential and its role in the ontology of the flow were developed by Bohm [credited: Bohm 1952]. The Bohm potential supplies the internal pressure that lets a localized matter wave *stand* rather than disperse, which is the established mechanism by which such a flow can be a self-trapped soliton. The FTGB spine (Part B) uses only the *exact* continuity half of this split; the identification of that continuity with helicity conservation is ours, but the split itself is textbook.

A.2.4 Topological solitons and baryon number as the pi_3(S^3) degree

A topological soliton is a smooth, finite-energy field configuration whose stability is guaranteed by a conserved integer – a topological *degree* – rather than by a balance of forces alone. Skyrme's model of the nucleon is the foundational example: a map from space (compactified to `S^3`) into the `SU(2)` group manifold (also `S^3`) carries an integer winding number, the degree of the map, which Skyrme identified with **baryon number** [credited: Skyrme 1961, *Proc. R. Soc. A* **260**, 127; Skyrme 1962, *Nucl. Phys.* **31**, 556]. The baryon number is the space integral of the topological current

```
B^mu = (1 / 24 pi^2) eps^{mu nu alpha beta} Tr( L_nu L_alpha L_beta ) ,   L_mu = U^{-1} d_mu U ,
```

and Witten established the physical interpretation of this current – and the quantization that makes the soliton a fermion – in the modern treatment [credited: Witten 1983, *Nucl. Phys. B* **223**, 422/433]. The crucial established point for this paper is that baryon number is a `pi_3(S^3)` degree, a *different* topological invariant on a *different* target space from the `pi_3(S^2)` Hopf helicity of Section A.2.2. The two are numerically independent: helicity is not baryon number. The higher-charge Skyrme solitons – in particular the `B = 4` cubic skyrmion identified with the alpha particle, and its classified vibrational spectrum – are established results [credited: Battye & Sutcliffe 1997, *Phys. Rev. Lett.* **79**, 363; Barnes, Baskerville & Turok 1997, *Phys. Rev. Lett.* **79**, 367] that the LENR boundary (Part E) draws on. Here they are noted only as standing physics.

A.2.5 Mass as a rate: the de Broglie / Compton identity

The final pillar is that mass has an exact reading as a *frequency.* De Broglie's foundational proposal attaches an internal periodic phenomenon – an internal clock – to every massive particle, with rest-frame angular frequency fixed by the Compton relation [credited: de Broglie 1924, *Recherches sur la theorie des quanta*],

```
hbar omega_C = m c^2     <=>     m = hbar omega_C / c^2 .
```

This is a dimensionally exact identity, not a model: it rewrites a mass in kilograms as the energy of a rate. The physical content of that internal clock – the "trembling" motion, or *zitterbewegung*, of the localized electron at `omega_zbw = 2 m c^2 / hbar`, twice the Compton rate – was given a concrete kinematic interpretation by Hestenes [credited: Hestenes 1990, *Found. Phys.* **20**, 1213]. The factor-of-two relation `omega_C = omega_zbw / 2` is the established origin of the clean 2:1 rotary structure that Part C reads as spin-1/2. As a further established consistency point, the charge/precession reading recovers the leading anomalous magnetic moment `a_e = alpha / (2 pi)` [credited: Schwinger 1948, *Phys. Rev.* **73**, 416]. In this work `m = hbar omega_C / c^2` is used strictly as this credited identity; the *interpretation* of the whirl frequency as the object's confined-flow rate, and the strict separation of the mass in kilograms from the dimensionless whirl number `q = 1/alpha`, are developed in Part C. No equation in this paper ever slides between a mass in kilograms and a pure number.

A.2.6 What Part A establishes

The five pillars above – force-free CK / Beltrami fields and their Woltjer-Taylor attractor; magnetic (and canonical) helicity as a conserved topological invariant; the exact Madelung / Bohm hydrodynamic form of quantum mechanics; Skyrme baryon number as the `pi_3(S^3)` degree; and mass as the rate `m = hbar omega_C / c^2` – are peer-reviewed, established physics. They are the load-bearing machinery of everything that follows. From here the paper builds the FTGB synthesis *on* this foundation: Part B states the one object and the organizing spine; Parts C and D read its structure and its living dynamics off that spine; and Part E carries the law to the nuclear boundary. Wherever a later claim is ours, it is tiered; wherever it rests on the physics of this part, that physics is credited here.

Part B. One object across scales

B.1 The spine: one force-free soliton, read at every scale

The synthesis rests on a single hypothesis: that a driven, force-free Beltrami-Hopf toroidal soliton is realized – at different sizes, windings, and coupling strengths – as the electron, the nucleon, the neutrino, and the Greenyer exotic vacuum object (EVO). "One object" is a claim about *form*, not about a single algorithm computing every property; Section B.3 states precisely where the readings share mathematics and where they diverge. Throughout, the object is read two ways at once: as a **field** configuration (a Chandrasekhar-Kendall force-free mode, `curl B = lambda B`) and as a **matter wave** in the Madelung-Bohm sense (`psi = sqrt(rho) e^{iS/hbar}`, the phase `S` carrying the flow). The equivalence of these two readings on this object is the medium<->matter law whose current leg is the subject of Part D; here we take the density leg (`rho = kappa (A.B)`, a profile proportionality [V]) as the working bridge.

The four anchors, zero structural free parameters. Given four measured inputs – a field strength `B`, a density `n`, a size `R`, and a species mass scale `m_i` – the dimensionless core of the object (Part C) is fixed with no further tunable structural knob. This is the sense in which the theory is "4 anchors / 0 free": the *shape* is determined; the anchors set the *scale*. (Parameters that look free – the mass-tower coefficient `a`, the beat depth – are treated honestly in Part C and the Honesty Ledger, Part F, as either derived-with-residual or [S] tautology-not-excluded.)

B.2 The family

- **Electron / lepton sector.** A chargeless-core helicity soliton dressed by a charged winding; mass set by the whirl (Compton) angular frequency (C.1). Charge is read as a spin-precession angular momentum (D4, QWM-tiered), not a primitive quantity.
- **Nucleon.** A Skyrme soliton of baryon degree `B = 1` (`pi_3(S^3)`, Witten 1983); the alpha particle is the `B = 4` cube (Battye-Sutcliffe 1997; Barnes-Baskerville-Turok 1997). Nuclear mass is the spin-precession whirl energy of the confined constituents; baryon number is the topological winding. These are distinct, and conflating them was a specific corrected error (mass-not-baryon).
- **Neutrino.** The pure force-free vortical ring – a chargeless, near-luminal "smoke-ring" (`pi_3(S^2)` helicity sector). It is naturally its own antiparticle: with no charge to conjugate, the Dirac/Majorana distinction maps onto the FTGB charged/chargeless split, and the smoke-ring is the *self-dual* (Majorana-like) member [S]. Neutrinoless double-beta decay is the external falsifier of that reading, not an FTGB prediction.
- **EVO.** A *charged*, non-relativistic electron cluster (micro-ball-lightning), needing the non-force-free charged sheath; distinct from the chargeless smoke-ring, and scoped to the Greenyer EVO specifically, not to universal ball lightning.

B.3 The two-sector split (stated honestly)

The unification is **partial by construction**, and the manuscript does not overclaim it. One substrate and one mass principle (`m = hbar omega / c^2`) run through every member, but two topological sectors require two matched computational methods:

- **Nuclear / baryon sector** – `pi_3(S^3)`, Skyrme baryon degree. Masses and reactions are computed by the **moduli-geodesic** method (soliton transition amplitudes; Part D, Toolkit M10). The many-body problem is replaced by a higher-degree soliton – no three-body-force tower – which is the sense in which the nuclear treatment is "beyond chiral EFT in *scope*" (never in *precision*; the Skyrme model is a model, Part F).
- **Lepton / EVO / neutrino sector** – `pi_3(S^2)`, Hopf helicity. Masses are read as **spectral eigenvalue-differences** of the Hopf-shell tower (C.2), not as moduli overlaps.

"One single algorithm for all sectors" is explicitly **rejected** as overreach: the substrate and the mass principle are shared; the computational machinery is sector-matched. This honest split is what keeps the synthesis a hypothesis rather than a slogan.

Part C. The derived core

C.1 Mass as a rate; charge as spin

Mass. The load-bearing identity is `m = hbar omega_C / c^2`, with `omega_C` the whirl (Compton) *angular* frequency [V, recovers `m_e` to machine precision; e/p/d rungs to <=0.32%]. This is the corrected form of the retired, dimensionally-wrong slogan “mass = whirl”: the whirl *number* `q = alpha^{-1} ~ 137` is a dimensionless ratio (orbital whirls per spin revolution), and mass is a *function* of it, `m = f(q)`, with the dimensional carrier `omega_C`. That `hbar omega_C / c^2` reduces to kilograms turns on `rad` being a dimensionless degree-of-freedom marker rather than a dimension – a point we hold obsessively (Toolkit; the dimensional re-audit is 8/8 clean). Mass is a rate: an obstruction to energy flow via spin-precession self-interference, not a Higgs coupling [QWM-tiered ontology; the SI-equivalent number is what is verified].

Charge. Read as a spin-precession angular momentum, `e ~ [kg rad/s]` (D4, QWM-tiered), which is what makes `m = e / omega` dimensionally consistent (SI’s current-times-time hides the relation by assigning charge its own dimension `[Q]`). The Kaluza-Klein fifth dimension, on this reading, is a topological charge from intrinsic spin precession – a reinterpretation, not a hidden spatial dimension (K1, tiered). No incumbent charge measurement is contested; the claim is an ontological re-reading with its own dimensional bookkeeping.

C.2 Topology and the mass tower

Topology. The *idealized closed-fibre reference field* carries integer Hopf invariant `Q_H = 1` (a credited property of the base Hopf map, used here only as an idealization label); the *actual CK object* carries a *real* helicity `H = 0.088` [V] (grid-convergent, sign-definite), NOT an integer Hopf charge – the CK field is not closed-fibre (the Whitehead pull-back is 54% non-solenoidal; `j_1(lambda R) = 0` does not compactify). So `Q_H = 1` is only the idealized-reference label, not the object’s integer, and `Q_H = B^2`-type identifications are excluded. In its photon/band structure the object carries Chern number `+/-2` [credited] / band `+/-1` [V-model]. Four distinct torsions (Cartan/Burgers, ribbon twist, Frenet, Ray-Singer) are kept separate; Einstein-Cartan torsion is excluded.

The mass tower [TUFT-preprint, tiered]. Nielsen’s Toroidal Unified Field Theory preprint supplies a spectral mass tower, `m_n ~ (n+1) exp(a n - zeta(3) n^2)`, with `a = 6 sqrt(2) exp(zeta(3)/24 pi^2) = 8.5285`. It reproduces the charged leptons to `0.0 / -0.4 / -2 %`. This is presented honestly as **tiered**: the coefficient `a` lands ~0.2% *off* the data optimum (a back-fit would hit it exactly), which is a genuine anti-fit signal, but the construction carries one free knob against one constraint, so its derivation status is `[S]`, tautology-not-excluded (Part F). The `n^2`-Casimir coefficients are pi-free by construction (`C5 = zeta(3)/12`); the `1/pi^2` normalization sits in a separate sector – a distinction we verified against the primary source after retracting an earlier misreading.

Koide. `K = (sum m)/(sum sqrt m)^2 = 0.66666051`, a `-9.2` ppm proximity to `2/3` that is non-generic (0 of 324 simple forms within 0.5%), with the trefoil `T(2,n)`, `n = 1,2,3`, mapping to the three lepton generations [category-pass]. This is `[V]` as a *relation on measured masses* and `[S]` as a *derivation* (one free knob); it is not claimed as derived.

C.3 The carrier comb and orbital angular momentum

The object’s spectral fingerprint is an **inharmonic carrier comb** built from the first CK Bessel roots (`4.4934 / 7.7253 / 10.9041`): frequencies `{121, 208, 294}` kHz in ratio `1 : 1.72 : 2.43` [V, spectroscopy]. The macroscopic whirl equals the comb fundamental (`omega_EVO = 7.6e5 = 2 pi * 121 kHz`, 0.035% – a promoted coincidence: the same collective mode). The inharmonicity is a discriminating prediction: a uniform or harmonic comb would falsify it. Its multibody phase-locking behaviour, and the sharpened comb-pull prediction (`f2/f1 -> 7/4`, `f3/f1 -> 5/2` under strong coupling), are treated in Part D. Orbital angular momentum steps by the octahedral order (`l -> l +/- N`), protected by the sheath winding; the QED-vacuum contribution to the effective refractive index is negligible.

End of Parts B-C. Load-bearing claims foregrounded on peer-reviewed work (Chandrasekhar-Kendall, Skyrme, *Witten, Battye-Sutcliffe, Barnes-Baskerville-Turok); Nielsen TUFT and QWM (Reed) marked tiered throughout; no over-unity; no number fabricated; ASCII-clean.*

Part D. The living dynamics and the current-leg trilogy

Parts B and C fixed the object's *shape* – the one object and its spine (Part B), with mass, charge, topology, and the carrier comb following from four measured anchors (Part C). This part sets that shape in *motion*, and then closes the one residual the shape left open.

Section D.1 shows the object is alive: not a parked eigenmode but a driven, dissipative limit cycle – a heartbeat – whose family self-organizes by textbook synchronization machinery, and whose collective rhythm carries a sharpened, falsifiable spectral prediction. Section D.2 is this session's strongest new result: the **medium-to-matter current-leg trilogy**, a tiered resolution of whether matter's flow can BE the plasmoid's own topological winding flux. It is a triple result – two proven no-gos, one constructed-realizable driven closure, a first-class selection negative, and two sharply named residuals – and is publishable on its own as a rigorous plasma / topological-fluid no-go-and-realizability theorem (see the closing note of D.2.8).

D.1 The toroidal-attractor beat generator

D.1.1 The object is alive: the Stuart-Landau heartbeat, not a flywheel

The static CK mode is only the shape. A real EVO is driven and dissipative: it persists by continuously trading energy with the polarizable medium, the way a candle flame or a convection cell persists – a **Prigogine dissipative structure**, renewed each cycle, not a spun-up flywheel [credited: Prigogine 1977]. The universal equation for such a self-oscillator sitting just past a Hopf bifurcation is the Stuart-Landau normal form. For supercritical drive the amplitude locks onto a ring of fixed radius `r*`, and the phase runs around that ring once per beat.

```
REAL-UNIT (amplitude dynamics)         |  Pi-GROUP (locked ring radius)
---------------------------------------|--------------------------------------
dw/dt = (mu + i*omega) w - beta|w|^2 w|  r* = sqrt(mu/beta) = sqrt(2)
mu = 1 (supercritical), beta = 1/2    |     = 1.414214   (rel. err ~1e-10)
                                       |  sub-critical mu = -0.3 -> decays to 0
---------------------------------------|--------------------------------------
Pi-group / tier: amplitude ratio |w|/r*; limit-cycle radius r* = sqrt(2) [V];
Stuart-Landau amplitude equation [credited: Stuart 1960 / Landau 1944].
```

Because the attractor is a *ring that never touches the origin*, the winding helicity `H ~ 0.088` rides the cycle conserved: the topology is protected by the dynamics, not merely parked. The physical realization is the **heartbeat** – a twist-writhe reconnection exchange that shuttles helicity between ribbon twist `Tw` and coil writhe `Wr` once per closure while the Calugareanu-White linking `Lk = Tw + Wr` is conserved [credited: Moffatt & Ricca 1992]. The object breathes at fixed radius `r* = sqrt(2)`, trading twist for writhe, without ever unwinding. This carries no over-unity: energy in equals energy dissipated each cycle. The same `r* = sqrt(2)` Lyapunov-stable ring re-appears in D.2.6 as the energy bound (X1) underwriting the current-leg global-existence residual – the heartbeat is what keeps the driven trajectory bounded for all time.

CLAIM -> Pi -> FALSIFIER. Claim: the object is a supercritical limit cycle. Pi: `|w| -> r* = sqrt(2)` regardless of initial amplitude. Falsifier: a driven EVO whose amplitude runs away or damps to zero (no stable finite ring) kills the dissipative-structure reading.

D.1.2 The beat generator: the toroidicity (TAE) gap law

The heartbeat is the *lowest* member of a spectral trichotomy that must be held apart throughout the paper: the eigen-comb (D.1.4, Part C.3), the torque harmonics (Part C.3), and the slow beat are three distinct objects, not three readings of one number. A toroidal Beltrami resonator does not have a single clean eigenfrequency; toroidicity splits neighbouring modes and opens a gap – exactly the toroidicity-induced Alfven eigenmode (TAE) gap of tokamak physics. The beat frequency is that gap, scaling *linearly* with the inverse aspect ratio to leading order in toroidicity.

```
REAL-UNIT (beat frequency)               |  Pi-GROUP (gap ratio)
-----------------------------------------|-------------------------------------
f_b = c_CK * eps * f_c                   |  Delta_omega/omega = f_b/f_c
    = c_CK * eps * v_A/(2 pi R)          |     ~= c_CK * eps   (LINEAR in eps)
f_c = v_A/(2 pi R)   (carrier/transit)   |  invariant gap (2D-axisym, eps->0):
eps = a/R_0 = 1/phi = 0.618 (canonical)  |     Delta_lambda = 0.208 * eps
eps_a = 0.697 (FreeFEM fatness)          |  FreeFEM 3D torus split at eps_a:
c_CK ~ 0.208 (near-universal 0.21-0.23)  |     Delta_lambda/lambda_1 = 0.0625 (~6%)
-----------------------------------------|-------------------------------------
Pi-group / tier: Delta_omega/omega with aspect eps; [S] (TAE Cheng-Chen-Chance 1985).
The gap is LINEAR in aspect ratio -- invariant gap Delta_lambda = 0.208*eps; the earlier
~eps^2 form was a carrier-normalization artifact (retracted, B8b). The FreeFEM 3D torus
split (0.0625, at FEM fatness eps_a = 0.697) is the higher-fidelity datum, in the ~6% band.
```

Two quantities must not be conflated with this gap ratio. First, the *observable modulation depth* is a separate, drive-set quantity of order 6-8% – set by how hard the object is driven, not by the geometry [S]. Second, an alternative "golden-rung" readout `f_b = f_2 - phi*f_1 ~ 0.102 f_1` gives `0.102`, differing by more than half from the derived toroidicity-gap value (~6%); it is a *different mechanism* and is carried strictly as a documented clue, never promoted [clue].

CLAIM -> Pi -> FALSIFIER. Claim: the beat is a toroidicity gap. Pi: `Delta_omega/omega` scales *linearly* in inverse aspect ratio (~ `eps`), vanishing as `eps -> 0`. Falsifier: a beat whose ratio is independent of `eps`, or scales nonlinearly (e.g. quadratically) in `eps`, falsifies the doublet-splitting (TAE) origin.

D.1.3 The Aizawa chaotic edge: a boundary, not a mechanism

As the drive strengthens, the same normal-form system passes near a chaotic neighbour of the Aizawa family – a bounded torus-with-a-hole with a positive largest Lyapunov exponent.

```
REAL-UNIT                               |  Pi-GROUP
----------------------------------------|-------------------------------------
lambda_max = 0.107  [1/time]            |  sign(lambda_max) > 0  =>  chaotic
(literature/toolkit band 0.10-0.124)    |  (the load-bearing claim is the SIGN)
----------------------------------------|-------------------------------------
tier: [V] positive-exponent detection (Benettin et al. 1980);
      Aizawa/Langford forced-Hopf normal form [credited]; FTGB reading [analogy].
```

The Aizawa branch is the **disordered neighbour** of the ordered heartbeat limit cycle – the *boundary* in parameter space between coherent beat-organization and decoherence, reached when the slow feedback strengthens. It is a boundary, not a mechanism: it makes no energy and does nothing nuclear. It is carried strictly as `[analogy]`; only the sign of the largest Lyapunov exponent is load-bearing.

D.1.4 The family self-organizes: Kuramoto / Adler / Arnold, on a KAM 3-torus

Because every family member is a driven limit cycle, a *collection* of them is a population of coupled phase oscillators – the setting of the standard, experimentally ubiquitous machinery of synchronization: Kuramoto for the many-body mean field, Adler for two-body locking, Arnold tongues for rational mode-locking. The onset of collective rhythm is governed by one dimensionless control parameter.

```
REAL-UNIT (dimensional threshold)         |  Pi-GROUP (control parameter)
------------------------------------------|-------------------------------------
Kuramoto:  theta_i"0027 = omega_i            |  P = K / K_c   (~ K / Delta_omega)
      + (K/N) sum_j sin(theta_j - theta_i) |  sync iff  P > 1
Adler (2-body): phi"0027 = Delta - K sin(phi)  |  numerical onset (N=800, Lorentzian):
K_c = 2/(pi g(0)) = 2*gamma (Lorentzian)   |     P ~ 1.2 at grid vs theory P = 1
                   = 1.596*sigma (Gaussian) |     (r ~ 0.04 -> 0.38 -> 0.71)
------------------------------------------|-------------------------------------
Pi-group / tier: order parameter r = |<exp(i theta)>|; threshold [credited: Kuramoto
1975/1984, Strogatz 2000]; numerical onset [V]; FTGB identification [S].
```

Applied to the three FTGB sub-populations, the single control `P = K/K_c` gives a modest, explicitly conditional picture (`K` is order-of-magnitude only, so every specific `P` is illustrative [S]):

subsystem	dimensional spread / K_c	control P	verdict	tier
Carrier comb {121, 208, 294} kHz	`Dw = 1.09e6, K_c ~ 6.9e5 rad/s`	`1.45` @ 4.3 nm; `~0.1` @ 10 nm	MARGINAL, coupling-limited	[S]
EVO e-cluster whirl `w_e = 7.76e20`	`Dw = 7.8e14..7.8e17`	`~1e-9..1e-12`	far BELOW, never locks	[S]
Deuteron seed (quantum)	control `n*lambda^3` vs Bose onset 2.612	`~0.01` warm; `1-4` cold seed	ABOVE at onset	[credited/S]

Two consequences are worth surfacing. The **electron internal whirl never locks, but the collective heartbeat does**: the whirl / Compton clock at `omega_C = 7.76e20 rad/s` (the zitterbewegung rate is `omega_zbw = 2 omega_C = 1.55e21 rad/s`) gives `P ~ 1e-9`, but the slow macroscopic collective whirl locks, closing a striking internal-consistency identity,

```
omega_EVO (macroscopic whirl)  =  2*pi * f_1(121 kHz)  =  7.603e5 rad/s   (0.035%)   [V]
```

– the family-axis whirl and the carrier-comb *fundamental* are the same clock (private fast clock, shared slow heartbeat). And the **deuteron seed independently re-derives the cold-seed requirement**: its synchronization variable is quantum phase coherence, with onset at Bose degeneracy `n*lambda^3 = 2.612`; a warm classical gas sits at `~0.01`, only the cold compressed seed (`n*lambda^3 ~ 1-4`) straddles onset. The synchronization view thus re-derives, from an entirely different direction, the cold-coherent-seed condition used in the LENR boundary (Part E.2) – a double cross-check, not an assumption.

Finally, the comb *protects its own rhythm.* Rational mode-locking occupies wedge-shaped Arnold tongues whose width scales as `K^q`, `q` the denominator of the rational approximation to a frequency ratio. The CK-Bessel comb ratios are stubbornly irrational – `f2/f1 = 1.719 ~ 43/25` (`q = 25`), `f3/f1 = 2.427 ~ 17/7` (`q = 7`), `f3/f2 = 1.413 ~ 24/17` (`q = 17`) – so for `K < 1` even the widest tongue is astronomically thin. Rather than collapsing onto one mode, the inharmonic comb lives on a quasiperiodic **KAM 3-torus**. Inharmonicity is a feature: it protects the multi-frequency rhythm from locking-collapse, and the choice `eps = 1/phi = 0.618` (the most-irrational aspect ratio) is the extreme case of that protection.

Throughout, "molecular organization" and "self-organization" mean **many-body oscillator organization only**. The defensible content is that a population of coupled nonlinear oscillators self-organizes once `K > K_c = 2/(pi g(0))`, a sharp transition verified numerically here [V] and experimentally ubiquitous (lasers, Josephson arrays, circadian cells, coupled metronomes). What is [S] is identifying *specific* FTGB-scale organization with that machinery. What is **not** claimed: no biological or chemical assembly, no energy gain, no nuclear rate. The word stays inside physics.

D.1.5 The sharpened comb-pull prediction [characterized]

The multibody analysis yields more than a picture: it sharpens the family's collective spectrum into a falsifiable prediction. A single family member on a KAM 3-torus resists locking; a *multibody* family couples through several harmonic channels at once, and the question becomes which rational – if any – the inharmonic carrier comb is pulled toward under realistic mode-coupling. This is decided by the *spectrum* of the harmonic couplings `k_h`, derived here from the plasmoid's own physics rather than assumed flat.

Three multiplicative factors set `k_h` relative to the fundamental: the perturbative order of the `h`-th temporal overtone (`eps^{h-1}`, credited weakly-nonlinear expansion); a geometric CK-Bessel triad factor `<C> = |V_111| = 2.30` [V, from the actual CK eigenmode overlaps]; and an inharmonic off-resonance factor `<D> ~ 0.86` [V] that *extra*-suppresses high overtones because integer multiples of the fundamental miss every CK eigenfrequency. Combined,

```
k_h / k_1 = alpha^(h-1),   alpha = eps * <C> * <D> = eps * 1.98.
```

The falloff is therefore **geometric (exponential in h)**, not power-law. The decisive constraint is self-consistency: a well-defined *comb* exists only in the weakly-nonlinear (sub-turbulent) regime `alpha < 1` (i.e. `eps < 0.51`); otherwise the perturbation series diverges and there is no clean line spectrum to lock. Hence for any physical comb `k_7/k_1 = alpha^6` is negligible (`1e-6 .. 1e-2` across the band), and the `h = 7` channel – the *only* channel that could pull the comb to the near-CK rationals `12/7` or `17/7` – is dead. (A power-law falloff would require a non-smooth sheath cusp; the smooth analytic plasmoid field gives exponential by Paley-Wiener. Flagged as the alternative that would change the answer.)

Feeding the derived falling `k_h` into a sine circle map with a robust plateau lock-test gives two regimes:

	realistic sub-critical coupling (`k1 <~ 0.6`, spacing `>~ 8-10 nm`)	strong near-field (`k1 >~ 0.8`, spacing `~4 nm`)
`f2/f1` (bare CK 1.719)	quasiperiodic, drifts near bare CK	locked to **7/4 = 1.750** (+1.8%)
`f3/f1` (bare CK 2.427)	quasiperiodic, drifts near bare CK	locked to **5/2 = 2.500** (+3.0%, quadratic)

The FTGB force-free / Lorentz dynamics is quadratic (3-wave) dominant, so `7/4 : 5/2` is the primary locked branch; a cubic (4-wave) drive would leave `f3/f1` unlocked or weakly at `12/5`.

PREDICTION (SHARPENED, falsifiable) [characterized]. An EVO / plasmoid near-field flux spectrum should show ONE of: (1) a quasiperiodic comb near the bare CK ratios `1.719 : 2.427` (incommensurate, no phase-lock – the realistic sub-critical outcome); OR (2) a mode-locked comb pulled to low rationals, `f2/f1 in [1.719, 1.750]` with attractor `7/4`, and `f3/f1 in [2.40, 2.50]` with attractor `5/2` – the strong-coupling outcome. **The sharp falsifier:** the comb will NOT be observed locked at the near-CK rationals `12/7 = 1.714` or `17/7 = 2.429`; those require the geometrically-dead `h = 7` channel, so a *measured* lock there would falsify the derived `k_h` spectrum. Coherence should also switch on/off as the near-field bead spacing crosses the `~4-10 nm` window that moves `K` across `K_c` (the D.1.4 Kuramoto test). The tier is `[characterized]`: the geometric falloff and the low-rational-or-no-lock dichotomy are robust; *which* of the two regimes obtains depends on the order-of-magnitude near-field coupling `k1`, so the prediction is a pinned two-branch band rather than a single ratio.

D.2 The medium-to-matter current-leg trilogy (this session's crown [V] result)

This section presents the paper's newest rigorous result. It concerns the central identification of the whole synthesis – that matter density and matter flow ARE the plasmoid's own topological winding density and flux – and settles its mathematical status as a tiered result: two proven no-gos, one constructed-realizable driven closure, a first-class selection negative, and two sharply named residuals, the deeper of which reduces to a recognized external open problem in fluid regularity.

The scope discipline (support-lens): the limits below are OUR computational limits on OUR object. Where a piece reduces to an external open problem (a regularity theorem), that is a statement of how far the theory has been driven, not a verdict against the phenomenon or against any collaborator's experiment.

D.2.0 The law, the two legs, and the density leg [V]

The FTGB object is a toroidal Beltrami plasmoid whose magnetic field `B` is force-free, `curl B = lambda B` (the field everywhere parallel to its own curl). The theory's central identification writes ordinary matter conservation as a *topological* continuity equation:

```
d_t rho + div(rho v) = 0          (ordinary matter conservation)
           ||
      d_mu K^mu = 0               (topological helicity current)
```

Here `K^mu = (K^0, K^i)` is the helicity 4-current: `K^0 = A . B` is the helicity *density* (the local linking / winding density of the field, `A` the vector potential) and `K^i` is its flux. The organizing law `rho = kappa (A . B)` says that where matter is dense, the field is densely wound. Two "legs" must hold for the full 4-current to close:

- **Density leg** – does `rho ~ K^0 = A . B` hold as a profile identity?
- **Current leg** – does the *flux* also match, `rho v ~ K^i`, so the full 4-current closes?

Density leg = [V]. In the Beltrami gauge `A = B/lambda`, the helicity density is `K^0 = A . B = |B|^2/lambda >= 0` (sign-definite, verified to `3e-15`). With the London relation `curl v = -(q/m) B` (a cold-condensate flow locked to the field, verified to `7e-15`), the matter-density profile tracks `|B|^2` with correlation `corr(rho, A.B) = 1.0000` – an exact dimensionless PROFILE identity. Crucially, `K^0` is a helicity density (units T^2 . m), not kg/m^3, so `kappa` is a genuine *dimensional* proportionality constant and the match is a profile proportionality, not a units identity [dimensional correction D2]. The density leg is settled. The whole subtlety – and the whole trilogy – lives in the current leg.

D.2.1 (a) The static no-go: closure requires |B| = const – [V]

Matching the densities forces the current *direction* too. London flow gives `v ~ A ~ B`, so the matter current is

```
rho v = (kappa |B|^2 / lambda) * (B / |B| scaling)  ~  |B|^2 B      (CUBIC in the field)
```

while the helicity flux is `K ~ B` (**LINEAR** in the field). Equality of a cubic and a linear quantity demands `|B|^2 = const`:

> **The 4-current closes if and only if `|B|` is constant.**

This was proven structurally and made basis-independent. The obstruction is measured by an operator residual: writing `L[phi] = curl((phi B - grad phi x B)/|B|^2)`, the residual is the distance of `B` from `range(L)`, which is dimensionless and `kappa`-independent – it lives in the current *direction*, so the D2 dimensional proportionality cannot absorb it. For the genuine FTGB Beltrami field the residual is `0.6053-0.6054` (basis-converged). A reducibility sweep of four physically motivated moves (gauge change, transverse E-field / sheath, norm choice, multi-lambda Beltrami) never reached the `< 0.2` reducible bar; in particular a transverse, non-force-free E-field leaves the residual UNCHANGED – the sheath CANNOT absorb it.

The deciding control is a null-free constant-`|B|` Beltrami field, `B = B0 (cos z, sin z, 0)`, which closes EXACTLY (residual `0.0000`). So the obstruction is entirely the spatial variation of `|B|` – and a topologically nontrivial closed-fibre Hopf / CK toroidal Beltrami field, which necessarily has field nulls, provably CANNOT have `|B| = const`. **The current leg is over-determined by construction – irreducibly a POSTULATE, not a theorem, in the static regime.** This closes the open item *positively*, as a first-class proven negative.

Physical narration. A real knotted field must thin out and vanish along some curves (the nulls of a nontrivial topology). Matter carried rigidly by that field would have to pile up and thin with `|B|^2`, but the winding flux it is supposed to equal only thins with `|B|`. The two book-keepings can agree only for a field that never thins – a field with no knot. Requiring "matter IS the flow of the knot" in a frozen field literally requires the knot to be untied. [Source: `CURRENTLEG_RESIDUAL_REDUCIBILITY`.]

D.2.2 Canonical helicity is the right conserved current – [V] / credited

One might hope a *driven* object, carrying its own conserved current, escapes the static theorem. It does carry one. The correct conserved current for a driven two-fluid plasma is **canonical (generalized) helicity**: with the canonical momentum `P = A + (m/q) v` and canonical vorticity `Omega = curl P = B + (m/q) curl v`, the current built from `H = integral P . Omega d^3x` is ideally conserved, `d_mu K_can^mu = 0`, *even when magnetic helicity is dissipated.* This holds because the canonical Ohm's law `E_can = -v x Omega` gives `E_can . Omega = 0` identically – canonical helicity is a Casimir invariant of the ideal two-fluid dynamics.

The prerequisite ("have your own conserved current first") is thus MET.

Tier and attribution. [V] / credited. The canonical-helicity-as-two-fluid-Casimir physics is carried by **Steinhauer & Ishida 1997** (PRL 79, 3423) and **Mahajan & Yoshida 1998** – these are the load-bearing citations. A third source – Bae, Kang & Shin, "On the double Beltrami states in Hall magnetohydrodynamics," arXiv:2504.07629 (2025) – independently classifies double-Beltrami states as energy minimizers under conservation of two helicities (a Casimir/variational result), corroborating the canonical-helicity Casimir. It is now **VERIFIED** as a real, on-topic paper; the primary load-bearing pair remains Steinhauer-Ishida 1997 + Mahajan-Yoshida 1998 (confirm exact Prop. label at lock). No physics claim in this section hangs on that single identifier. [Source: `DRIVEN_CANONICAL_HELICITY_4CURRENT`.]

D.2.3 (b) The aligned/steady no-go: the (A'B'C')^2 >= 0 obstruction – [V], adversarially verified

Having the right conserved current does not, by itself, close the current leg – it only *relocates* the obstruction. The canonical flux is `K_can = h v + (mu - P.v) Omega` (with `h` the canonical helicity density and `mu` the Bernoulli head), so matter-current closure `(rho, rho v) ~ (h, K_can)` requires `(mu - P.v) Omega = 0`. On a double-Beltrami equilibrium the Bernoulli head is constant, `mu = const`, so this becomes the exact structural mirror of the static theorem:

> **Aligned / steady closure requires `P . v = const`** (the analog of `|B| = const`).

This second leg was proven a genuine no-go and then *adversarially* verified. The crux: making the relevant cross-term vanish forces the two Beltrami parameters to be equal and opposite, `lambda_- = -lambda_+`; then `P.v = const` requires a POSITIVE ratio `p1^2 / p2^2 > 0` between two ABC helicity-fluctuation triples. This is impossible, because

```
(A"0027B"0027)(B"0027C"0027)(A"0027C"0027) = (A"0027B"0027C"0027)^2 >= 0
```

is a perfect square, so the required constancy can be met only by driving the field to a topologically TRIVIAL state. Independent verification: the residual reaches zero only at `c = -0.604 < 0`, the forbidden-sign branch. Distinct `lambda` values force a spectral band-gap, hence single-mode collapse – and the spectrum is discrete on ANY compact domain, so there is no continuous-CK loophole. The physical corner was upgraded `[S] -> [V]-via-mechanism`: `std(P.v)/mean` equals the field-magnitude inhomogeneity to `~1%` (for close `lambda`, `Omega ~ lambda P` gives `P.v ~ lambda |P|^2`, which collapses back onto the static `|B| = const` theorem). No 3D closer exists – the lowest residual `0.198` only trivializes, while the genuine FTGB object sits at `~0.58`. The two [V] no-gos are thereby unified: they are the same obstruction, seen once in the magnetic field and once in the canonical momentum.

[Sources: `DRIVEN_PV_CONST_NOGO_PROOF` + `DRIVEN_PV_CONST_NOGO_VERIFY`.]

D.2.4 (c) The driven resolution: closure <=> S = 0, realizable by an electrostatic drive – [S leaning V]

The third leg differs *in kind* – it is NOT a no-go. Define the scalar `S = mu - P.v`. Because `S` is a SCALAR, the closure condition `S Omega = 0` forces `S = 0` pointwise: there is no "S perpendicular to Omega" vector branch (an earlier misreading, verified false to `4e-17`). The canonical current is then ALWAYS a matter current `(h, h u)` with `u = v + (S/h) Omega`, closing on the fluid velocity `v` exactly when `S = 0`.

Two facts make this a genuine resolution rather than a third obstruction.

(1) The S = 0 closure surface is non-empty and drive-compatible. Pointwise the surface is NON-EMPTY and compatible with a bounded physical drive (residual `0.00`; jointly satisfiable to `~1e-16`) – categorically UNLIKE the two no-gos, whose closure surfaces are EMPTY. A degree-of-freedom count makes the distinction sharp: the isolated ideal system is 8/8; adding closure makes it 9/8 (over-determined -> the two no-gos), but a sustained external DRIVE supplies exactly the one spare freedom the driven regime has. [Source: `DRIVEN_NONALIGNED_CLOSURE`.]

(2) The `f_ext perp Omega` constraint was a body-force ARTIFACT. In an earlier framing, maintaining `S = 0` seemed to require an extra constraint `f_ext . Omega = 0` on the drive. The 3D-realizability analysis showed this was an artifact of representing the drive as a *mechanical body force*. The exact identity is `d_mu K_can^mu = 2 f_body . Omega`; a purely *electromagnetic* (Lorentz + barotropic) force has `f_body = 0`, so `d_mu K_can^mu = 0` for ANY `E . B` – the canonical-helicity Casimir again. The `S = 0` drive is then simply the electrostatic potential

```
phi = A . v + (m / 2q) |v|^2 - w/q
```

with one constraint and one knob, giving residual zero (`S ~ 3e-16`, `E.B ~ 1.4-2.5 != 0`, genuinely non-aligned). The differential-algebraic system is **index-1** (`dS/dphi = 1 != 0`): `phi` is algebraically slaved, there is no diverging drive, and it stays bounded on the heartbeat trajectory. The two-species closure `{S_e = 0, S_i = 0}` is NOT re-over-determined – the scalar potential `phi` and the parallel potential `A_parallel` (along the current `v_e - v_i`) absorb both, with Jacobian `det = -|v_e - v_i|^2`, invertible if and only if current flows. **Tier: [S] leaning [V].** [Source: `CURRENTLEG_3D_REALIZABILITY`; 1D precursor `CURRENTLEG_TORSION_TRANSPORT_REALIZABILITY`.]

Physical narration. The frozen field could not carry matter as its own winding because the two book-keepings scaled differently. But a *driven* object – one with a live electric field doing work along the field lines (`E . B != 0`), i.e. the real heartbeat plasmoid rather than a frozen idealization – has a spare degree of freedom, an electrostatic potential it can set, that exactly reconciles the flow with the winding flux. Matter *can* be the flow of the knot, but only in a knot that is being actively driven.

D.2.5 (d) The selection negative: the object's own dynamics select the no-go – [V]/[S] NEGATIVE

A crucial honesty question remains: does the object CHOOSE `S = 0` on its own, or must it be externally tuned there? The answer is a specific, tested NEGATIVE – `S = 0` is TUNED, not self-selected, and the object's own dynamics select AGAINST closure.

- **(A) Variational.** Minimizing energy at fixed canonical helicities (Woltjer / Taylor / Mahajan-Yoshida Euler-Lagrange) yields double-Beltrami equilibria with Bernoulli head `mu = const`, hence `S = const - P.v != 0`. The natural extremum sits MAXIMALLY FAR from `S = 0`. The only functional that `S = 0` extremizes is `integral S^2 |Omega|^2` – the closure residual squared, a tautology (gate FAIL). Montgomery-Phillips minimum-dissipation and cross-helicity relaxation likewise give `mu = const`.
- **(B) Dynamical.** A driven-dissipative Hall two-fluid integration (spectral RK4; two initial conditions x {relax, Taylor-drive, force-drive with `E.B != 0`}) keeps `rms(S)/dPv` at O(1) (1.3-2.2, growing in two runs); the driven attractor RELAXES TOWARD the aligned `mu = const` no-go (`sin(v, Omega)` falling 0.80 -> 0.17 and 0.74 -> 0.39). The **Stuart-Landau heartbeat limit cycle of D.1.1 sits OFF the S = 0 surface and ON the no-go** (genuineness defect `1.4e-3`).

Tier [V]/[S] NEGATIVE – a specific, tested negative, NOT a category dismissal: a tuned external drive still realizes closure (D.2.4); the object simply does not select it. As a bonus, the static and aligned no-gos are thereby shown to be the *dynamical attractor* of relaxation, reinforcing them. **Consequence:** closure is externally DRIVEN, never emergent. "Matter is knotted flow" is barred as a frozen, self-organizing identity; it is available only as a driven, externally-tuned construction. [Source: `SELECTION_PRINCIPLE_S0`.]

D.2.6 The residuals: R1 benign, R2 <=> 3D enstrophy / BKM – [S] conditional

The 3D-realizability construction left two sharply named residuals standing between [S leaning V] and full [V].

R1 – the current-null `v_e = v_i` degeneracy. The two-species drive scales as `1/|v_e - v_i|` and so diverges where the electron and ion currents coincide (the two-species analog of the static `|B| = const` null theorem). This is resolved as **genuine-but-BENIGN [S]:** the set `{v_e = v_i}` is codimension-3 – isolated POINTS, coincident with the CK / Hopf B-nulls – and hence MEASURE ZERO (contrast the static `|B| = const`, a finite-VOLUME killer). The drive is singular pointwise (`alpha ~ C/delta^2`) but the fields `E, B` stay bounded and the drive energy is `L1 + L2` integrable, so the crossings contribute nothing to any integral; closure holds on the full-measure current-full complement. [Source: `CURRENTLEG_R1_CURRENTNULL`.]

R2 – global all-time existence on a bounded, current-full Maxwell-two-fluid trajectory. This is reduced to **[S] CONDITIONAL:** R2 holds provided three ingredients, and R2 <=> X3 given X1, X2:

- **X1** – energy / `L^2` bound: **HAVE**, via the Stuart-Landau heartbeat Lyapunov function `dV/dt <= 0` at `r* = sqrt(2)` (the same limit cycle of D.1.1). Canonical helicity bounds the energy from *below* – heartbeat persistence – which is the *wrong* direction for blow-up.
- **X2** – no current-null crossing: **HAVE**, via R1.
- **X3** – the enstrophy / `H^1` bound, i.e. the Beale-Kato-Majda condition `integral ||Omega||_inf dt < inf`: **OPEN**.

X3 is category-IDENTICAL to open large-data 3D Navier-Stokes / Hall-MHD global regularity (**Chae-Degond- Liu 2014**, Ann. IHP C 31, 555; Beale-Kato-Majda 1984, CMP 94, 61) – a recognized *external* open PDE problem, NOT an FTGB-specific gap. Numerically the driven system is stable to horizon `T = 6000` (about 2196 heartbeat periods): `|S| <= 2.2e-16`, drive bounded throughout – but this horizon is a LOW-MODE reduced ODE that cannot exhibit the small-scale enstrophy blow-up by construction, and the object is high-Lundquist (`S ~ 1e2-1e6`), hence super-critical; the result is therefore weak evidence, NOT sub-criticality. R2 stays [S] hard-open. [Source: `CURRENTLEG_R2_GLOBAL_EXISTENCE`.]

D.2.7 The current leg, resolved (honest tier)

Assembling the trilogy:

> **The medium-to-matter current leg is a [V] no-go / well-founded POSTULATE in the frozen regimes** (static `|B| = const` and aligned/steady `P.v = const` – both proven, both adversarially or mechanism-verified), **and a CONSTRUCTED-REALIZABLE closure [S leaning V] in the genuinely-driven regime**, via an external electrostatic + circuit drive, obstructed only on a measure-zero benign null set. **The object's own variational and attractor dynamics SELECT the no-go** (a first-class NEGATIVE): closure is externally driven, not emergent. **The sole remaining gate to full [V] is the 3D enstrophy / BKM regularity bound – a recognized external problem, not an FTGB gap.** The density leg stays [V]; canonical helicity is a genuine ideal conserved current [V]/credited.

This resolves the "one named residual" (ledger L1) that Part B.1 flagged when it first stated the spine. The residual is not erased – it is *located*: everything internal to the theory is closed, and the one edge that remains is a famous problem in fluid regularity, driven to, not stumbled into.

D.2.8 Standalone-publishability note

The current-leg trilogy is publishable *on its own*, independent of the wider synthesis, as a rigorous plasma / topological-fluid result: a proven no-go (matter-current closure for a force-free field requires `|B| = const`), its canonical-helicity generalization (aligned closure requires `P.v = const`, obstructed by the perfect-square identity `(A"0027B"0027C"0027)^2 >= 0`), a constructive electrostatic-drive realizability theorem for the genuinely time-dependent regime, a selection negative showing the equilibrium and relaxation dynamics both avoid the closure surface, and a clean reduction of the sole remaining gate to the Beale-Kato-Majda enstrophy bound for 3D Hall-MHD. It stands as the strongest single [V] contribution of the program and is the natural carve-out should a full-synthesis venue prove too broad (see the architecture's venue note).

End of Part D. No locked file edited; no number fabricated; canonical-helicity load-bearing citations = Steinhauer-Ishida 1997 + Mahajan-Yoshida 1998 (Bae-Kang-Shin 2025 now VERIFIED real, corroborating); the 3D-BKM reduction cites Chae-Degond-Liu 2014 + Beale-Kato-Majda 1984. No over-unity: the heartbeat is a Prigogine dissipative structure, energy in equals energy dissipated each cycle. ASCII-clean throughout.

Part E – The nuclear boundary

Parts A through D built the object and its living dynamics: one driven, force-free Beltrami-Hopf standing wave, read at once as a Chandrasekhar-Kendall plasma mode and as a Madelung matter wave, organized by the spine `rho = kappa (A.B)` and its current-leg trilogy. Part E turns outward. It states, first, exactly where this reading sits relative to the incumbent physics it is built on (E.1), and second, what the object does and does not claim at the low-energy-nuclear boundary (E.2). Both sections are governed by one discipline: the account adds no number that established physics already fixes, and every extension carries its own falsifier. Limits reported here are our computational limits on our object; where a piece reduces to an external open problem, that records how far the theory has been driven, not a verdict against any experiment or collaborator.

E.1 Domain of validity: five pillars vs the Standard Model and classical EM

The object-reading does not overturn the Standard Model or classical electromagnetism. It is a coherent-object *re-reading* laid atop tested physics at a small number of specific joints, plus one genuine extension (a conserved topological helicity current) that Maxwell permits but does not require. This section makes that relationship precise, pillar by pillar. For each of the five load- bearing pillars – mass, charge, the wavefunction, electromagnetism, and the nuclear sector – we state the incumbent account (credited exactly where it is tested and load-bearing), the FTGB re- reading (a specific statement, never a dismissal), and the specific testable bound that separates them or, more often, that shows them agreeing on every measured number.

Two disciplines govern the whole table and are stated once here so they need not be repeated in each row. First, **FTGB adds no number the incumbent already fixes.** The whirl reading returns the electron rest energy `m_e c^2` to six figures; the charge reading returns the Schwinger anomaly `a_e = alpha/(2 pi)`; the Madelung reading returns quantum continuity to machine precision (`~5e-16`); and quantum chromodynamics is left entirely intact. Second, **each re-reading carries its own falsifier** (right column), so each extension is decidable rather than decorative. FTGB is a re-reading and an extension of tested physics at four specific joints, and it leaves QCD untouched.

PILLAR	INCUMBENT (credited where tested)	FTGB RE-READING
MASS	Standard Model: mass from the Higgs Yukawa coupling, fit and confirmed at the electroweak scale (LHC).	m = hbar omega_C / c^2 -- mass is the energy of the internal whirl rate; the whirl NUMBER q = 1/alpha is separate and dimensionless.
CHARGE	QED: charge a primitive gauge quantum number; g-2 confirmed to ~1e-12.	charge = mechanical spin angular momentum e ~ [kg.rad/s], a Cartan/ Burgers loop-closure torsion defect.

WAVEFUNCTION	Born rule: psi a probability amplitude; ubiquitously confirmed.	Madelung flow: psi = sqrt(rho)exp(iS/h) read as a real compressible fluid; the Bohm potential lets the wave stand.
EM	Maxwell: exact and tested across every accessible regime.	Maxwell + a conserved topological helicity current d_mu K^mu = 0; the QHD reading treats the same fields as a matter flow.
NUCLEAR	QCD: confirmed by lattice + deep-inelastic scattering; the deuteron / He-4 sector measured.	QCD untouched; the open CMNS anomaly is re-read as scaffold-not-reaction -- environment + formation + orientation + disposal supplied, kernel inherited.

The nuclear row is the one that carries the most weight in a synthesis paper and the one most easily misread, so it is worth stating in words. FTGB proposes no modification to the strong interaction: the confinement scale, the deuteron and He-4 structure, and the measured cross-sections of standard nuclear physics all stand exactly as the incumbent leaves them. What FTGB re-reads is not the nuclear force but the *low-energy anomaly* reported in the condensed-matter-nuclear-science (CMNS) literature – and it re-reads that anomaly as a question about environment, formation, orientation, and disposal, with the sub-barrier kernel treated as an inherited, well-posed open computation rather than a new principle. Section E.2 makes that division exact.

None of the five re-readings is offered as a refutation of the incumbent it sits beside. Each column of established physics is credited precisely where it is tested; each re-reading is a specific, falsifiable statement about the *same* tested numbers seen through a coherent-object ontology. The domain of validity of FTGB is therefore not a rival domain to the Standard Model or Maxwell: it is a reading laid over them, decidable at four joints and open at one (the nuclear kernel), which is the subject of the rest of Part E.

E.2 The LENR anomaly and the kernel boundary: scaffold-not-reaction

E.2.1 The headline, stated positively

Carried to the nuclear scale, the medium-matter law does not become a new nuclear force. It becomes a sharp, decidable statement about *which parts* of the low-energy-nuclear (LENR / CMNS) anomaly the driven Beltrami-Hopf object **supplies** and which parts it **inherits**. This is the theory's positive nuclear headline: **scaffold-not-reaction.** The electron-cluster plasmoid (the EVO) builds the room in which the reaction becomes possible and orients the players inside it; it does not itself push two deuterons through their mutual Coulomb wall. Each side of the boundary is a well-posed calculation, not a missing principle.

Throughout, the CMNS observables are accepted as reported without re-adjudicating them (support-lens): He-4 correlated with heat at roughly 24 MeV per helium atom (Miles 1993, replicated), a replicated loading threshold with COP approximately 1.3-1.4 – which is **not** over-unity (McKubre 1994), small transmutation yields (Iwamura 2002), and gammaless heat (Storms 2007) [credited / contested]. These are treated as accepted-not-adjudicated inputs; the limits reported below are ours, not verdicts on those experiments. Following the field's citation hygiene, we do not cite E-Cat / Rossi, hydrino / Mills, or biological-transmutation claims [do-not-cite].

The structure of the section is: the environment the object supplies is verified (E.2.2); the "frequency desert" objection is closed on five independent kinetic-delivery routes AND shown to be decoupled from the actual open number (E.2.3); the experimental anomalies are structurally accounted for without any frequency mechanism (E.2.4); and the quantitative closure reduces to exactly two static inputs with two different owners – an FTGB-internal branching amplitude and a host-lattice- inherited screening energy (E.2.5-E.2.7). FTGB derives no COP and no rate; every such number is flagged as inherited or open.

E.2.2 What the object supplies: the verified environment, formation, orientation, and disposal

The scaffold side of the boundary is the part the object genuinely supplies, and it is the part the verified core of the theory already delivers.

Environment [V/S]. The CMNS literature infers a coherent, cold, degenerate host under three names – the nuclear-active environment (NAE, Storms), the tetrahedral symmetric condensate (TSC, Takahashi), and the coherent-correlated state (CCS, Vysotskii). The driven, charged Beltrami-Hopf EVO of Parts B-D *is* that host: one object read three ways. The upstream chain is the verified core of the paper – an EVO cascade condensing into the plasmoid, carrying the inharmonic CK carrier comb `{121, 208, 294} kHz` and a coherent multi-body population that self-organizes by the Kuramoto / Adler / Arnold dynamics of Part D.

Formation, derived cold [V/S]. The force-free bulk together with the non-force-free charged theta-pinch sheath assembles deuterons *cold*: degenerate-sea Landau drag *removes* kinetic energy from an infalling ion rather than adding it, so the sheath is a cooling, sorting boundary, not a heater. Species-sorting raises the local density roughly 10-100x, carrying the deuteron phase-space occupancy across the Bose-degeneracy threshold `n lambda_dB^3 ~ 1-4` (onset ~2.612), while the confinement kinetic energy per ion sits at `~1e-4 .. 1e-3 eV` – five to six orders below any barrier scale [V energy floor; S sorting]. A multibody Kuramoto cross-check re-derives the same `n lambda_dB^3 >= 2.612` degeneracy gate from the family-synchronization side, an independent double-check of the cold-seed condition.

Orientation and the aneutronic channel [S]. The clean channel `d + d -> He-4` is read as a topological merger of windings, `B = 2 + B = 2 -> B = 4`, into the octahedral (cube) Skyrmion ground state (Battye & Sutcliffe 1997). Helicity `K^0 = A.B` – a real number `H ~ 0.088` for the CK object – *orients* the merger, selecting the relative alignment on the cold path; baryon number `B` is conserved independently by `d_mu B^mu = 0` [credited]. Baryon number is conserved throughout this section.

Disposal character [S]. The 23.847 MeV released leaves by a mechanical fissility / ejecta continuum – the near-BPS `B = 4` configuration sheds energy into a kinetic ejecta continuum that does possess states near 20 MeV, so the classic "no 20 MeV mode" objection does not apply to the continuum channel. Aneutronicity is dynamical (the cold trajectory selects the He-4 sheet while the `n + He-3` and `p + t` exits remain symmetry-allowed but kinematically suppressed), and gammalessness is a genuine selection rule: the `0+ -> 0+` transition from the `S=0, L=0` entrance to the He-4 ground state is single-photon E0-forbidden [credited: Church-Weneser 1956]. No fixed formation energy enters the disposal accounting.

E.2.3 The frequency desert: closed on five routes, and then shown irrelevant

The hardest-looking objection to any account in which a macroscopic object participates in a nuclear event is the **frequency desert.** The EVO's macroscopic clocks – the kHz-MHz drive, the ~27 kHz heartbeat, the carrier comb at 121-294 kHz, and even the whirl at `~7e16` Hz – are separated from the nuclear / relative coordinate (`~1e18 .. 1e24` rad/s) by a gate of roughly 12-19 orders of magnitude. Every *kinetic* route by which drive energy might be delivered into the deuteron-deuteron relative coordinate must bridge that gate. Five independent kinetic-delivery routes were computed and closed:

1. **Direct single / superharmonic pump** – requires a harmonic order `n ~ 1e12 .. 1e15`. DEAD [V].
2. **Internal resonance / autoresonance** – evades the ponderomotive common-mode zero, but the single-step commensurability is `p/q ~ 1e14` and chirp capture bridges less than one order of the gap. DEAD [V]/[S]. (Autoresonance is the fifth route, grouped here with internal resonance.)
3. **Broadband self-similar (MHD-turbulence) up-cascade** – terminates 9-13 orders short of the nuclear coordinate; per-mode energy `~1e-23 .. 1e-37` J and phase-randomized – the structural opposite of phase-matched. NO [V].

4. **Parametric / seesaw frequency down/up-conversion** – the rung count is modest (~30-70 rungs), so rungs are not the obstruction. The killer is the disposal identity generalized to the up direction: for virtual / off-resonant rungs,

   `eta^N = (r^N)^-2 = R^-2   for EVERY N`

   (verified for `N = 1, 5, 20, 60` over a 12-order gap, all equal to `1e-24`). Subdividing a virtual jump gains nothing; the result is direction- and N-independent. A cascade is real only with genuine on-shell resonances at `eta ~ O(1)`, and the one *measured* electromagnetic/atomic-to-nuclear anchor – NEET (Nuclear Excitation by Electron Transition, on-resonance by construction, i.e. the most favorable case) – gives `P < 4.1e-10` (189-Os; Kishimoto 2006) [exact bound to verify at lock], 9-14 orders below O(1). Real coherent up-conversion technology (high-harmonic generation) spans only ~2.5-3 orders. NEGATIVE, category-distinct.

Alongside these five kinetic routes, a *separate* penetration-side track (object-sourced coherent- correlated build-up) is closed by three independent negatives – a wrong-clock frequency mismatch to the ~135 GHz local coordinate, the empirical Vysotskii apparatus ceiling (`r ~ 0.98`), and local electron-bath decoherence 5-7 orders faster than accumulation. This penetration-side track is not one of the five kinetic routes; it is why only the mainstream, measured-but-contested screening `U_s` survives as an inherited input rather than an object-derived one (see E.2.6).

The decoupling: the desert is a red herring for the resolution. The decisive move is not another bridge but a *decoupling.* The one genuinely open kernel number is `Delta = M_fi(theta)`, a **static nuclear-structure matrix element**, and the theory's accepted formation mechanism is cold and structural (compression + screening + cold Landau-Zener branching), not a heated frequency cascade. Two disciplined shortcuts confirm the decoupling by refutation: the internal-clock (Compton/whirl) degeneracy between identical deuterons is a pure global c-number phase that cancels in any transition probability [REJECT]; and Bose-degenerate bunching supplies only a ~2x encounter-rate prefactor [credited but insufficient]. **LENR resolution therefore does not require bridging the frequency desert. The desert is genuinely un-bridgeable (five closed routes, [V]/NEGATIVE) AND genuinely irrelevant: the aneutronic-branching anomaly reduces exactly to computing one static B=4 reactive overlap, decoupled from the desert by construction.**

E.2.4 The experimental anomalies, structurally accounted – without any frequency mechanism

Because the resolution is decoupled from the desert, the reported anomalies are already accounted for structurally, independent of any frequency-delivery route:

- **Aneutronic excess heat / missing neutrons and gammas.** Cold formation avoids the hot `p+t` and `n+3He` compound; the `0+ -> 0+` E0 transition is single-photon-forbidden ("no hard gamma") [credited: Church-Weneser 1956]; and a dynamical cold Landau-Zener step selects the aneutronic direction [S].
- **He-4 / heat correlation at ~24 MeV.** The Q-value is exact: `Q(d+d -> He-4) = 23.847 MeV` (CODATA / AME2020, two independent routes agreeing to <0.01 MeV) [V]. Disposal is a mechanical ejecta continuum – fractionation over `N ~ 1e2 .. 1e4` fragments to 1-100 keV each, a many-soft-few-fast, gammaless signature [S].
- **COP approximately 1.3-1.4.** This is a scaffold enhancing an ordinary aneutronic exit, not over-unity. **FTGB derives NO COP.** The 1.3-1.4 figure is inherited field positioning; any device / COP framing is excised-as-refuted. The heartbeat of Part D is a Prigogine dissipative structure, not a flywheel; helicity rides conserved and no free energy is produced.

E.2.5 The kernel as a well-posed matrix element and its factorization

The inherited kernel is not a vague gap; it is a specific, published-grounded open nuclear-structure computation. In the standard framework `M_fi = <Psi_f | O | Psi_i>`, `Gamma = (2 pi/hbar) |M_fi|^2 rho_f`, the entrance state is the antisymmetrized two-cluster state of two `B=2` toroidal deuteron Skyrmions with Coulomb-distorted relative motion; the exit state is the `B=4` octahedral cube (He-4, 0+) dressed with its own published vibrational spectrum (Barnes-Baskerville-Turok 1997; Gudnason-Halcrow 2018); and the operator is a collective (monopole / quadrupole / octupole) transition operator of the standard Bohr-Mottelson kind evaluated on the soliton's own fluctuation eigenbasis [credited structure; S application].

The branching amplitude factorizes cleanly by channel `alpha` (partial-wave / multipole):

```
M_fi(theta) = Sum_alpha  C_alpha(theta)  A_alpha  R_alpha  S_alpha
```

- `C_alpha(theta)` – orientation / angular geometry (spherical harmonics, rotation matrices). **SETTLED [V].**
- `A_alpha` – angular-momentum coupling (Clebsch-Gordan / Wigner-Eckart). **SETTLED [V].**
- `S_alpha` – spin / isospin / spectroscopic / antisymmetrization / many-body structure. [**S**], on the published B=4 collective-mode basis.
- `R_alpha` – the radial / reduced overlap integrals with their relative phases. **OPEN** [**S**] – the ab-initio NCSMC / RGM piece, the deciding number.

A from-scratch `O_h` character table (self-checked against Tinkham / Koster) settled `C_alpha` and `A_alpha` [V]: the reaction is confined to the gerade sector, odd-parity octupole (breakup-ward) modes are parity-forbidden from the `0+` s-wave entrance, and the `0+ -> 0+` gamma is E0-suppressed. But the **`n+3He`** and **`p+t`** exits are symmetry-allowed, so the symmetry factor is at most `~O(10)` – seven to nine orders short of the `~1e8 .. 1e9` needed. Two independent derivations (a desk refutation and the character table) agree: aneutronic dominance is **dynamical / kinematic** (a cold Landau-Zener avoidance), not a symmetry rule.

E.2.6 The two-input closure: two numbers, two different owners

Quantitative LENR closure requires **two static inputs, with different owners.** Distinguishing them is the sharpest statement of scaffold-not-reaction, and both are stated here without fabricating any value.

Input 1 – Delta (branching): FTGB-internal, open [S]. `Delta` is the off-diagonal `B=4` two-diabatic-surface reactive Landau-Zener gap – the amplitude between the bound `B=4` cube and the `B=3 + B=1` breakup surface, meeting at a crossing, with suppression `S = exp(delta), delta = pi Delta^2 / (2 hbar v |dF|)`, exponentially sensitive to `Delta`. This is the aneutronic- branching / anomaly number, and it is FTGB's own to compute. It is not pinned in-band [S]. The blocker is diagnosed and priced rather than hand-waved: both tractable ansaetze fail at the crossing (the product ansatz hits the repulsive core wall; naive field-blending breaks topology, `B -> 2.4`), and a genuine 3D Skyrme relaxation of the *massless* energy unwinds the topology (`|B| -> 0`) at every affordable lattice spacing. The unblocking path is a named, priced external computation: a topology- preserving discretization (or `dx <= 0.06-0.10 fm`, `N >= 100-170`) plus a pion-mass term plus a compiled / GPU arrested-Newton minimizer plus two-branch continuum matching – estimated `~15 h/sweep` minimum, days at robust resolution, genuinely intractable in-budget and reported as a blocker, not dressed as an answer [label / BLOCKED; A].

Input 2 – U__s (penetration / screening): host-lattice inherited, accepted-not-adjudicated. The sub-barrier penetration/screening energy `U_s ~ 300-800 eV` (Czerski 2001; Raiola 2002/2004; well above the Assenbaum-Langanke-Rolfs 1987 theory floor of ~25 eV) is measured-but-contested and **inherited from the field, not FTGB-derived.** An explicit attempt to derive it from the object's own degenerate electron density fails: Thomas-Fermi screening on the FTGB density gives only `U_s ~ 18-39 eV`, i.e. the mainstream theory floor – the *same side* of the gap the field itself cannot cross. This is a first-class NEGATIVE: FTGB cannot source the anomalous screening, and, tellingly, the anomaly appears in *plain metal lattices* independent of any EVO, so `U_s` is properly the host lattice's inherited, unresolved question, accepted-not-adjudicated here.

The two inputs have **different owners**, which is the crux: `Delta` (branching) is FTGB-internal and open; `U_s` (penetration) is host-lattice-inherited and contested. The kinetic-delivery side is closed on five routes; the aneutronic-branching anomaly reduces exactly to `Delta`; and full *rate* closure additionally needs `U_s`, which stays inherited-open. The anomaly is not a category impossibility; it is a well-posed open number plus one inherited screening input.

E.2.7 The executed diagonal, and what it is NOT

To locate `Delta` honestly it is worth stating what *has* been executed. The diagonal `2 x (B=2) -> B=4` merger quantum was computed on real 3D Skyrme field integrals (a genuine attractive well at `sigma* ~ 2.5 fm`, the inertia `mu(sigma)` calibrated to `M_d/2`, quantized by the same period-integral operator validated on the sine-Gordon breather to `1.4e-6`), giving `hbar omega_merger = 66-95 MeV` (two constructions x two resolutions), with a two-sided-corrected floor of ~30 MeV – tens of MeV. This diagonal quantum is order-consistent with the primary-source BBT "cube -> two donuts" merger mode (~20 MeV; Barnes-Baskerville-Turok 1997), a genuine two-operator convergence (executed moduli-geodesic diagonal versus BBT Hessian `Eg`).

Three cautions are load-bearing and stated explicitly:

1. The executed diagonal **refutes** the earlier 1.33 MeV static-ratio identification (it differs by 20-50x); that number is not the gap.
2. The diagonal quantum is **not the branching gap**: setting `Delta := hbar omega_merger` in `S = exp(delta)` gives `log10 S ~ 9600`, the freeze-both-channels pathology – positive proof that the diagonal is the wrong object. The branching amplitude is the *off-diagonal* `Delta` of E.2.6, which remains open [S].
3. The diagonal ~20-30+ MeV also reproduces the measured `4He 0+_2` level at 20.21 MeV (Bacca 2015), but this is a **soft, not a clean, cross-check**: the `0+_2` is a `p+t` breakup Feshbach resonance (`Gamma ~ 270-500 keV`), not a clean bound state, so it is an order-consistency check only, not clean corroboration. The genuine result is the BBT ~20 MeV two-operator convergence.

Tier for `V, mu, hbar omega_merger`: **[S-model, EXECUTED].** No nuclear rate, cross-section, or gap magnitude is fabricated anywhere in this section.

E.2.8 Summary and the open frontier

The object supplies the environment (NAE / TSC / CCS as one host), the cold formation (`KE ~ 1e-4 .. 1e-3 eV`, occupancy `n lambda_dB^3 ~ 1-4`), the merger orientation (helicity-oriented `B=4` octahedral merger), and the aneutronic, baryon-conserving disposal character (dynamical aneutronicity + a genuine E0 `0+ -> 0+` gamma ban + mechanical fissility into a continuum). The frequency desert is closed on five kinetic-delivery routes and shown decoupled from the resolution. The experimental anomalies – aneutronic excess heat, He-4/heat at ~24 MeV, COP 1.3-1.4, and missing gammas and neutrons – are structurally accounted for by formation, the exact Q-value, the E0 selection rule, and a dynamical Landau-Zener step, with no frequency mechanism required and no COP derived.

Quantitative closure reduces to two static inputs with two owners: `Delta`, the off-diagonal `B=4` reactive Landau-Zener gap (FTGB-internal, open [S], its blocker priced as a named external Skyrme computation), and `U_s`, the sub-barrier screening energy (host-lattice inherited, accepted-not- adjudicated; FTGB's own density gives only ~18-39 eV, and the anomaly appears in plain metal lattices). Its selection skeleton `{C_alpha, A_alpha}` is settled [V]; its collective structure `S_alpha` is [S] on a published basis; its diagonal energy scale is executed [S-model, ~20-30+ MeV] and cross-checks against BBT. The 1.33 MeV static ratio is refuted as the gap; the true branching `Delta` is not pinned in-band [S]. Scaffold-not-reaction is a measured boundary, not a dismissal: FTGB supplies the scaffold, inherits one contested screening number from the host lattice, and leaves one well-posed branching amplitude as its own named external computation. No over-unity is claimed; no rate, cross-section, or gap magnitude is fabricated; baryon number is conserved throughout.

Part F. Honest limitations and the open frontier

This section states the *real, existing* problems of the working core – the places where the theory is genuinely a postulate, a model, or an inherited-not-derived input – and nothing else. Exploratory coincidence-avenues (numerical near-hits pursued and set aside because they did not survive a genuine mechanism test) are kept in the working record but are omitted here: they do not bear on the reproducible positive core, and a paper is not a lab notebook. What follows are limitations that a referee would rightly raise, disclosed plainly.

F.1 The medium-to-matter current leg is a postulate, not a full theorem

The density leg of the law (`rho = kappa (A.B)`, a profile proportionality) is `[V]` for this object. The **current** leg is not a theorem, and we say so. In the static and aligned/steady regimes it is a proven `[V]` no-go: the matter 4-current closes only if a field magnitude is spatially constant (`|B| = const`, then `P.v = const`), which a topologically nontrivial (nulled/knotted) field cannot satisfy – so the law stands there as a consistent *defining postulate*, not a derived identity. In the genuinely-driven regime the closure is *constructed-realizable* (`[S]` leaning `[V]`) by an external electrostatic drive, but the object's own variational and attractor dynamics **select the no-go**: closure is externally driven, never emergent. This is an honest structural limit of the law, not a hidden assumption.

F.2 Four open inputs that reduce to named external problems

Each of the following is a real gap in the working core. None is an internal contradiction; each reduces to a *specific, named, external* computation or ingredient – which is the honest shape of the theory's present boundary.

- **The LENR branching amplitude `Delta`** (off-diagonal `B=4` two-diabatic-surface Landau-Zener gap) is `[S]`/open. It is a static nuclear-structure matrix element, decoupled from the frequency desert; its computation is a named external task – a topology-preserving, pion-massive, fine-grid Skyrme relaxation (the reduced ansaetze provably fail in the crossing region; the blocker is diagnosed and priced, not hand-waved). The executed *diagonal* mode (~20 MeV, cross-checked to the `4He 0+_2` level) is confirmed but is not the branching gap.
- **The screening energy `U_s`** (~300-800 eV) is a real second input to any LENR rate, and it is **inherited, not FTGB-derived**: the object's own degenerate electron density yields only ~18-39 eV (the mainstream floor), and the enhanced screening is observed in ordinary metal lattices with no plasmoid present – it is host-lattice metallurgy (accepted-not-adjudicated), owned by the material, not by the object.
- **The full-`[V]` current-leg realizability** reduces, via a clean chain, to a **3D enstrophy / Beale-Kato-Majda regularity bound** – category-identical to the recognized open large-data global-regularity problem for 3D Navier-Stokes / Hall-MHD, a famous *external* problem, not an FTGB-specific gap.
- **The neutrino mass hierarchy** is not derived by the object's spectrum; closing it requires an **external Majorana (lepton-number-violating) seesaw sector**. The theory does supply, honestly, the generation *count* (trefoil `T(2,n)`), the oscillation *mechanism* (whirl-beat), and a mixing *clue* (`mu-tau` symmetry forcing `theta_23 ~ 45 deg`) – but not the mass values.

F.3 The honest pattern

The theory has been driven to its boundaries, and the boundaries share a clean shape: every deepest open input reduces to a **named external computation or ingredient** – 3D-BKM regularity, a massive-Skyrme relaxation, host-lattice screening, a Majorana seesaw sector – rather than to an internal contradiction or a fabrication. Two further core claims carry explicit caveats a referee should see: the spectral mass tower is `[TUFT-preprint]`-tiered with a one-knob coefficient (tautology-not-excluded), and the Koide relation is `[V]` on measured masses but `[S]` as a derivation. Stating these limits precisely is what lets the positive core (Parts A-D) stand as a reproducible object rather than an overclaim.

End of Part F. Real existing problems only; exploratory not-found trails omitted from the publication (retained in the working record); no number fabricated; ASCII-clean.

Part G. Methods and references

G.1 Methods and toolkit

The synthesis is supported by two reusable method modules, stated here in compact form; each carries its own tier and validity note.

G.1.1 M9 – The coupled-oscillator, helical, topologically-constrained substrate

FTGB is one member of the established class of driven-dissipative nonlinear coupled oscillators on a helical, elastic, topologically-constrained substrate. The mapping is tiered as **7 IDENTITY** rows (same mathematics, already load-bearing in FTGB: the Stuart-Landau normal form as the heartbeat with `r* = sqrt(2)`; Kuramoto/Adler/Arnold as the multibody beat-locking; `Lk = Tw + Wr` and the four torsions as the object's topology; the Madelung phase as the flow reading), **5 ANALOG** rows (shared mathematics, different substrate – DNA chiral-rod torsion transport, BZ chemical oscillators, biological entrainment, the phonon / Pd-D lattice), and **4 CAUTION** rows (imported as discipline: vibration is not a single cross-scale coherence; classical oscillation is not de Broglie coherence; coherent matter-wave chemistry is an ultracold regime; "molecular" means many-body organization, not assembly/energy/nuclear claims). It is a **synthesis, not an identity claim**: no biopolymer, chemical, or solid-state system is asserted to *be* an FTGB object.

G.1.2 M10 – Topological-soliton methods

Five methods, tiered:

1. **Canonical / generalized helicity conserved current** [V/credited]. `H = integral P.Omega`, `P_s = A + (m_s/q_s) v_s`, `Omega_s = curl P_s`; a Casimir of the ideal two-fluid dynamics, ideally conserved even as magnetic helicity dissipates. Full statement and the `E"0027.Omega = 0` derivation are given in Part D.2.2; carried on Steinhauer-Ishida 1997 and Mahajan-Yoshida 1998.
2. **No-go template for magnitude-inhomogeneity obstructions** [V]. A conserved-current closure holds iff a field magnitude is constant; impossible for nontrivial (nulled/knotted) topology. The scalar reduction `S = mu - P.v` (closure iff `S = 0`; there is no vector "`S perp Omega`" branch); the `(A"0027B"0027C"0027)^2 >= 0` positivity obstruction; and the measure-zero (benign, isolated nulls) vs finite-volume (killer) classification.
3. **Moduli-geodesic transition-amplitude method** [S-model, sine-Gordon-validated]. Collective coordinate, metric `mu(sigma)`, potential `V(sigma)`, geodesic/Landau-Zener amplitude; validated in 1+1 sine-Gordon (kink mass to `1.4e-6`, kink-antikink force to 0.9%, breather scaling `1.93` vs `2.0`) with an honest factor-~2.2 ceiling on near-threshold coefficients. **Load-bearing usage warning:** the DIAGONAL `(mu,V)` vibrational quantum is NOT the OFF-DIAGONAL branching gap (they differ by orders; the `S = exp(delta)` freeze-both gate proves it); executing the off-diagonal amplitude requires real relaxation, not a reduced ansatz. Do not dress a static energy ratio as a geodesic amplitude.
4. **Seesaw frequency-downconversion** [S] / [V-dim]. Via `m = hbar omega / c^2`, the seesaw `m_light ~ m_D^2 / M_R` reads as `omega_light ~ omega_D^2 / omega_R`, a parametric-beat downconversion; the whirl ladder is a calibrated mass<->frequency map (`omega_nu -> 0.05` eV, 0.04%), and a genuine seesaw requires an off-ladder scale.
5. **RG dielectric-flow / IR-fixed-point method** [S-mechanism] / [flagged value]. `K_PV(q^2) = alpha(0)/alpha(q^2)`; running-coupling and static-winding as the same object at two scales; 137 as the IR endpoint; with a genericity-denominator anti-numerology control (count how many simple invariants land within a tolerance before crediting any single one).

G.2 References

Load-bearing FTGB claims are foregrounded on peer-reviewed work; the Nielsen TUFT preprint and Reed QWM are marked tiered wherever used and are never cited as established. Established-convergence vs FTGB-novel is labelled per use. Flags carried below are binding at lock.

Force-free fields, helicity, topology. Chandrasekhar & Kendall 1957, ApJ 126, 457; Woltjer 1958, PNAS

44, 489; Taylor 1974, PRL 33, 1139; Moffatt 1969, JFM 35, 117; Moffatt & Ricca 1992, Proc. R. Soc. A 439, 411; Etnyre & Ghrist 2000, Nonlinearity 13, 441; Calugareanu-White-Fuller (`Lk = Tw + Wr`).

Two-fluid canonical helicity (driven current-leg, [V/credited]). Steinhauer & Ishida 1997, PRL 79, 3423; Mahajan & Yoshida 1998, PRL 81, 4863. Bae, Kang & Shin 2025, "On the double Beltrami states in Hall magnetohydrodynamics," arXiv:2504.07629 – **VERIFIED real and on-topic** (double-Beltrami states classified as energy minimizers under conservation of two helicities, a Casimir/variational result); corroborates the canonical-helicity Casimir. The primary load-bearing pair remains the two references above (confirm exact Prop. label at lock).

Hydrodynamic / matter-wave QM. Madelung 1927, Z. Phys. 40, 322; Bohm 1952, Phys. Rev. 85, 166; de Broglie 1924; Hestenes 1990, Found. Phys. 20, 1213.

Skyrme solitons and nuclei. Skyrme 1961/1962, Proc. R. Soc. A 260, 127 / Nucl. Phys. 31, 556; Witten 1983, Nucl. Phys. B 223; Battye & Sutcliffe 1997; Barnes, Baskerville & Turok 1997, PRL 79, 367 (B=4 16-mode spectrum); Houghton-Manton-Sutcliffe 1998, Nucl. Phys. B 510, 507; Feist, Lau & Manton 2013, PRD 87, 085034; Gudnason & Halcrow 2018, PRD 98, 125010; Halcrow 2016, Nucl. Phys. B 904, 106; Manton, "Classical Skyrmions" (arXiv:1106.1298); Adam, Sanchez-Guillen & Wereszczynski 2010, PLB 691, 105 (BPS Skyrme).

Ab-initio nuclear structure. Hupin, Quaglioni & Navratil 2019, Nat. Commun. 10, 351 (d+t -> n+4He NCSMC benchmark); Quaglioni & Navratil 2008, PRL 101, 092501; Navratil & Quaglioni 2012, PRL 108, 042503. *Corrections carried:* Navratil-Quaglioni 2011 (d+4He NCSMC, PRL 106) is an A=6 (6Li) system, not A=4; Bacca et al. 2015, PRC 91, 024303 – the 4He 0+_2 (20.21 MeV) is a p+t breakup Feshbach resonance (`Gamma ~ 270-500` keV), so a SOFT/order-consistency cross-check, not a clean bound-state corroboration. Church & Weneser 1956, Phys. Rev. 103, 1035 (E0 gamma suppression).

3D regularity (external open problem). Beale-Kato-Majda 1984, CMP 94, 61; Chae, Degond & Liu 2014, Ann. IHP C 31, 555 (Hall-MHD small-data global / large-data local).

Nuclear excitation by electron transition (desert-bridge negative anchor). Kishimoto et al. 2006, PRC 74, 031301(R) (189Os, `P < 4.1e-10` – *exact bound to re-verify at lock*); 197Au, PRL 85, 1831 (2000).

Neutrino sector (external). PDG-2024 neutrino-mixing review + NuFIT 5.2; Centelles Chulia et al. 2024, JHEP 07 (2024) 060 (arXiv:2404.15415, Type-I seesaw family); PRD 100, 075029 (2019); canonical Minkowski / Gell-Mann-Ramond-Slansky / Yanagida / Mohapatra-Senjanovic seesaw; Schechter-Valle.

Screening (host-lattice, accepted-not-adjudicated). Czerski et al. 2001; Raiola et al., Eur. Phys. J. A; Huke, Czerski & Heide 2008, PRC 78, 015803; Assenbaum-Langanke-Rolfs 1987; Ichimaru (dense-plasma screening review).

Coupled-oscillator analog domains (M9, established cross-domain). DNA torsional-stress transport (Nelson 1999, PNAS 96, 14342; and Nat. Commun. 16, 10543, 2025); BZ mechano-chemical resonance (PNAS 2320331121); multiscale biological oscillators (PMC12855917); multi-channel entrainment (Cell Systems, 2023).

Do-not-cite (per field knowledge base): Rossi, Mills, and biological-transmutation (Kervran) claims are excluded; the biophysics references above concern ordinary torsion/oscillator physics and are unrelated to "biological transmutation."

End of Part G. Every citation real and verified; flags binding at lock (Bae-Kang-Shin now verified real; Bacca caveat; Navratil-Quaglioni-2011 = A=6; Kishimoto bound re-verify); established vs FTGB-novel labelled; ASCII-clean.